

PRINCIPLES OF FINANCE/INTRODUCTION TO FINANCE

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TOPIC ONE: NATURE AND SCOPE OF FINANCE

Required Rate of Return (R_i)

The required rate of return (R_i) is the minimum rate of return that a project must generate if it has to receive funds. It's therefore the opportunity cost of capital or returns expected from the second best alternative. In general,

Required Rate of Return = Risk-free rate + Risk premium

Risk free rate is compensation for time and is made up of the real rate of return (R_r) and the inflation premium (IR_p). The risk premium is compensation for risk of financial actions reflecting:

The riskiness of the securities caused by term to maturity

The security marketability or liquidity

The effect of exchange rate fluctuations on the security, etc.

The required rate of return can therefore be expressed as follows:

$$R_j = R_r + IR_p + DR_p + MR_p + LR_p + ER_p + SR_p + OR_p.$$

Where:

R_r is the real rate of return that compensate investors for giving up the use of their funds in an inflation free and risk free market.

IR_p is the Inflation Risk Premium which compensates the investor for the decrease in purchasing power of money caused by inflation.

DR_p is the Default Risk Premium which compensates the investor for the possibility that users of funds would be unable to repay the debts.

MR_p is the Maturity Risk Premium which compensates for the term to maturity.

LR_p is the Liquidity Risk Premium which compensates the investor for the possibility that the securities given are not easily marketable (or convertible to cash).

ER_p is the Exchange Risk Premium which compensates the investors for the fluctuation in exchange rate. This is mainly important if the funds are denominated in foreign currencies.

SR_p is the Sovereign Risk Premium which compensates the investors for the possibility of political instability in the country in which the funds have been provided.

- OR_p is the Other Risk Premium e.g. the type of product, the type of market, etc.

The risk can be divided into two:

- a. The diversifiable (unsystematic) risk;
- b. The non-diversifiable (systematic) risk.

The diversifiable risk is that risk which the investor can be able to eliminate if he held an efficient portfolio.

The non-diversifiable risk on the other hand is those risks that still exist in all well diversified efficient portfolios. The investor therefore seeks to eliminate the diversifiable risk. This can be shown below:

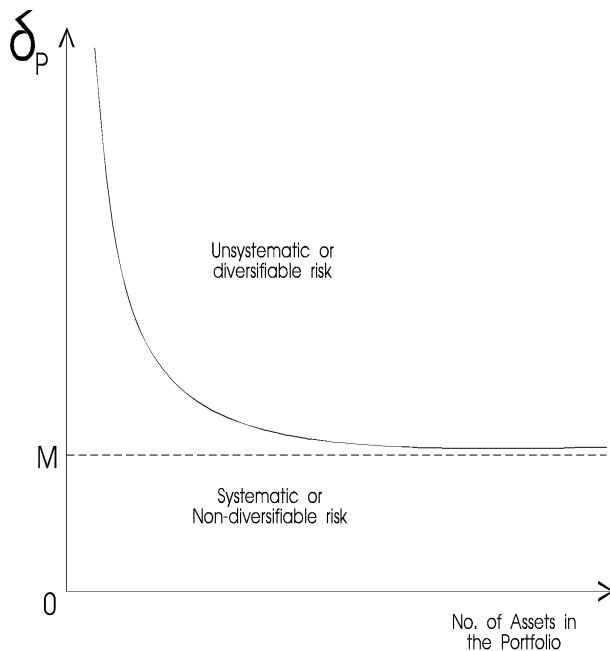


Figure 1 *Diversification of Risk*

From the graph shown above as the number of assets increases, the portfolio risk reduces up to point M. At this point the lowest risk has been achieved and adding more assets to the portfolio will not reduce the portfolio risk. An efficient portfolio therefore is well diversified portfolio.

Note: The non-diversifiable risk can also be referred to as the market risk.

Scope/Elements

1. Investment decisions includes investment in fixed assets (called as capital budgeting). Investment in current assets are also a part of investment decisions called as working capital decisions.
2. Financial decisions - They relate to the raising of finance from various resources which will depend upon decision on type of source, period of financing, cost of financing and the returns thereby.
3. Liquidity decision- The firm's liquidity refers to its ability to meet its current obligations as and when they fall due. It can also be referred as current assets management. Investment in current assets affects the firm's liquidity, profitability and risk. The more current assets a firm has, the more liquid it is. This implies that the firm has a lower risk of becoming insolvent but since current assets are non-earning assets the profitability of the firm will be low. The converse will hold true. The finance manager should develop sound techniques of managing current assets to ensure that neither insufficient nor unnecessary funds are invested in current assets.
4. Dividend decision - The finance manager has to take decision with regards to the net profit distribution. Net profits are generally divided into two:

1. Dividend for shareholders- Dividend and the rate of it has to be decided.
1. Retained profits- Amount of retained profits has to be finalized which will depend upon expansion and diversification plans of the enterprise.

Objectives of Financial Management

The financial management is generally concerned with procurement, allocation and control of financial resources of a concern. The objectives can be-

1. To ensure regular and adequate supply of funds to the concern.
2. To ensure adequate returns to the shareholders which will depend upon the earning capacity, market price of the share, expectations of the shareholders.
3. To ensure optimum funds utilization. Once the funds are procured, they should be utilized in maximum possible way at least cost.
4. To ensure safety on investment, i.e, funds should be invested in safe ventures so that adequate rate of return can be achieved.
5. To plan a sound capital structure-There should be sound and fair composition of capital so that a balance is maintained between debt and equity capital.

Functions of Financial Management

1. **Estimation of capital requirements:** A finance manager has to make estimation with regards to capital requirements of the company. This will depend upon expected costs and profits and future programmes and policies of a concern. Estimations have to be made in an adequate manner which increases earning capacity of enterprise.
2. **Determination of capital composition:** Once the estimation have been made, the capital structure have to be decided. This involves short- term and long- term debt equity analysis. This will depend upon the proportion of equity capital a company is possessing and additional funds which have to be raised from outside parties.
3. **Choice of sources of funds:** For additional funds to be procured, a company has many choices like-
 1. Issue of shares and debentures
 1. Loans to be taken from banks and financial institutions
 1. Public deposits to be drawn like in form of bonds.

Choice of factor will depend on relative merits and demerits of each source and period of financing.

4. **Investment of funds:** The finance manager has to decide to allocate funds into profitable ventures so that there is safety on investment and regular returns is possible.
5. **Disposal of surplus:** The net profits decision have to be made by the finance manager. This can be done in two ways:
 1. Dividend declaration - It includes identifying the rate of dividends and other benefits like bonus.
 1. Retained profits - The volume has to be decided which will depend upon expansion, innovational, diversification plans of the company.

6. **Management of cash:** Finance manager has to make decisions with regards to cash management. Cash is required for many purposes like payment of wages and salaries, payment of electricity and water bills, payment to creditors, meeting current liabilities, maintenance of enough stock, purchase of raw materials, etc.

Financial controls: The finance manager has not only to plan, procure and utilize the funds but he also has to exercise control over finances. This can be done through many techniques like ratio analysis, financial forecasting, cost and profit

TOPIC 2: FINANCIAL PLANNING/ FORECASTING

Financial forecasting refers to determination of financial requirements of the firm in advance. This requires financial planning using budgets. The financial planning and forecasting will also determine the activities the firm should undertake in order to achieve its financial targets.

Financial forecasting is important in the following ways:

1. Facilitate financial planning i.e. determination of cash surplus or deficit that are likely to occur in future.
2. Facilitate control of expenditure. This will minimize wastage of financial resources in order to achieve financial targets.
3. It avoids surprise to the managers e.g. any cash deficit is known well in advance thus the firm can plan for sources of short term funds such as bank drafts or short term loans.
4. Motivation to the employees - Financial forecasting using budgets and targets will enhance unity of purpose and objectives among employees who are determined to achieve the set target.

Methods/Techniques of Financial Forecasting

1. Use of Cash Budgets

A cash budget is a financial statement indicating:

- a) Sources of revenue and capital cash inflows
- b) How the inflows are expended to meet revenue and capital expenditure of the firm.
- c) Any anticipated cash deficit/surplus at any point during forecasting period.

CASH BUDGETING

Parts of the Cash Budget

- In a simple cash budget, there are three parts:
 1. The Worksheet Area
 2. The Inflows and Outflows
 3. The calculation of the ending cash balance
- Essentially, a cash budget starts with the beginning cash balance, adds expected cash inflows, and subtracts any expected cash outflows. The result is the expected ending cash balance.
- A cash budget is a document which shows the expected cash inflows and outflows for a chosen time period (say, 6 or 9 months).
- The benefits of the cash budget are:

3. It provides an estimate of the ending cash balance in each month
3. It provides estimates and sources of the cash inflows and outflows
3. It provides a basis of comparison against which managers can be evaluated

Cash Budget: An Example

1. Here is the information required to assemble the as follows:

	April	May	June	July	August	September	October
Sales	Shs 291,000	365,000	387,000	329,000	238,000	145,000	92,000

2. 40% of sales are for cash. Of the remainder, 75% is collected the following month, and 25% is collected two months after the sale.
3. Inventory purchases are equal to 50% of the next month's sales. 60% of purchases are paid for the following month, and the remainder one month later.
4. Wages are 20% of sales. Leasing expenses is Kshs10,000 per month. Interest payments of Kshs 30,000 are due in June and September. A Kshs 50,000 dividend payment will be made in June. Taxes of Kshs 25,000 are due in June and September. A Kshs 200,000 capital improvement will be paid for in July.
5. The company must keep a minimum cash balance of Kshs 15,000.

Cash Budget: Worksheet Area

The worksheet area is where we gather certain key numbers which will be used in the rest of the cash budget.

Bithlo Barbeques Cash Budget For the Period June to September 1998								
	April	May	June	July	August	September	October	
Sales	291,000	365,000	387,000	329,000	238,000	145,000	92,000	
<i>Collections:</i>								
Cash	40%		154,800	131,600	95,200	58,000		
First Month	45%		164,250	174,150	148,050	107,100		
Second Month	15%		43,650	54,750	58,050	49,350		
Total Collections			362,700	360,500	301,300	214,450		
Purchases	50%	182,500	193,500	164,500	119,000	72,500	46,000	
<i>Payments:</i>								
First Month	60%		116,100	98,700	71,400	43,500		
Second Month	40%		73,000	77,400	65,800	47,600		
Total Payments			189,100	176,100	137,200	91,100		

Cash Budget: Inflows & Outflows

This section shows all of the cash collections and disbursements. Note that these are only cash inflows and outflows. The cash budget is not the same as the income statement.

Collections		362,700	360,500	301,300	214,450
<i>Less Disbursements:</i>					
Inventory Payments		189,100	176,100	137,200	91,100
Wages	20%	77,400	65,800	47,600	29,000
Lease Payment		10,000	10,000	10,000	10,000
Interest		30,000	0	0	30,000
Dividend (Common)		50,000	0	0	0
Taxes		25,000	0	0	25,000
Capital Outlays		0	200,000	0	0
Total Disbursement		381,500	451,900	194,800	185,100

Cash Budget: The Ending Cash Balance

This section is where we calculate the ending cash balance and determine if we will need to borrow for each month.

Beginning Cash Balance		20,000	15,000	15,000	121,500
Collections - Disbursement		(18,800)	(91,400)	106,500	29,350
Unadjusted Cash Balance	20,000	1,200	(76,400)	121,500	150,850
Current Borrowing Needed	0	13,800	91,400	0	0
Ending Cash Balance	20,000	15,000	15,000	121,500	150,850
Notes:					
6.	Minimum Acceptable Cash 15,000				

2. Regression Analysis

This is a statistical method which involves identification of dependant and independent variable to form a regression equation $y = a + bx$ on which forecasting will be based on.

3. Percentage of Sales Method

This method involves expressing various balance sheet items that are directly related to sales as a percentage of sales.

Assumptions underlying % of sales method

The fundamental assumption underlying the use of % of sales method is that, there is no inflation in the economy i.e. the increase in sales is caused by increase in production and not increase in selling price.

Other assumptions include:

1. The firm is operating at full or 100% capacity. Therefore the increase in production will require acquisition of new fixed assets.

2. The firm will not issue new ordinary shares or debenture or preference shares thus this capital will remain constant during the forecasting period.
3. The relationship between balance sheet item and sales i.e balance sheet items as % of sales will be maintained during forecasting period.
4. The after tax, profit on sale or net profit margin will be achieved and shall remain constant during the forecasting period.

It involves the following steps:

- i) Identify various balance sheet items that are directly with sales this items include:
 - a) Net fixed asset - If the current production capacity of the firm is full an increase in sales will require acquisition of new assets e.g. machinery to increase production.
 - b) Current Asset - An increase in sales due to increased production will lead to increase in stock of raw materials, finished goods and work in progress. Increased credit sales will increase debtors while more cash will be required to buy more raw materials in cash.
 - c) Current liabilities - Increased sales will lead to purchase of more raw materials
 - d) Retained earnings - This will increase with sales if and only if, the firm is operating profitability and all net profits are not paid out as dividend.

Note:

The increase in sales does not require an increase in ordinary share capital, preference share capital and debentures since long term capital is used to finance long term project.

- ii) Express the various balance sheet items varying with sales as percentage of sales e.g. assume for year 2002 stock and net fixed assets amount to Sh.12M and 18M respectively sales amount to Sh.40M. Therefore stock as percentage of sales"

$$\text{Stock} =$$

$$\text{Fixed asset} =$$

- iii) Determine the increase in total asset as a result of increase in sales e.g suppose sales increases from Sh.40 M to Sh.60 M during year 2003. The additional stock and net fixed asset required would be determined as follows:

$$\text{Increase in stock} = \quad \% \text{ of sales} \times \text{increase in sales}$$

$$= 30\% (60 - 40) = \text{Sh.6M}$$

$$\text{Increase in fixed asset} = \quad \% \text{ of sales} \times \text{increase in sales}$$

$$= 45\%(60 - 40) = \text{Sh.9 M}$$

- iv) Determine the total increase in assets which will be financed by:

- a) Spontaneous source of finance i.e increase in current liabilities

Where Increase = % of sales x increase in sales

- b) Retained earnings for the forecasting period
Retained earnings = Net profit - Dividend paid

$$\text{Net profit margin} = \frac{\text{Net profit}}{\text{Sales}}$$

Therefore: Net profit = Net profit margin (%) x sales

Note:

Generally Net profit margin is called after tax return on sales.

Out of the total assets that are required as a result of increase in sales, the financing will come from the two sources identified. Any amount that cannot be met from the two sources will be borrowed externally on short term basis which will be a current liability.

Illustration

The following is the balance sheet of XYZ Ltd as at 31st December 2021:

	Sh.'000'
Net fixed asset	300
Current assets	<u>100</u>
	<u>400</u>
Financed by:	
Ordinary share capital	100
Retained earnings	70
10% debentures	150
Trade creditors	50
Accrued expenses	<u>30</u>
	<u>400</u>

Additional Information

1. The sales for year 2021 amounted to Sh.500,000. The sales will increase by 15% during year 2022 and 10% during year 2023.
2. The after tax return on sales is 12% which shall be maintained in future.
3. The company's dividend payout ratio is 80%. This will be maintained during forecasting period.
4. Any additional financing from external sources will be affected through the issue of commercial paper by company.

Required

- a) Determine the amount of external finance for 2 years up to 31st December 2023
- b) Prepare a proforma balance as at 31 December 2023

Solution

Identify various items in balance sheet directly with sales:

Fixed Asset

Current Asset

Trade creditors

Accrued expenses

$$\text{Net fixed assets} = \frac{300\text{M}}{500\text{M}} \times 100 = 60\%$$

$$\text{Current assets} = \frac{100\text{M}}{500\text{M}} \times 100 = 20\%$$

$$\text{Trade creditors} = \frac{50}{500} \times 100 = 10\%$$

$$\text{Accrued expenses} = \frac{30}{500} \times 100 = 6\%$$

c) Compute the increase in sales over the 2 years.

$$\text{Year 2003 sales} =$$

$$\text{Year 2004 sales} =$$

$$\text{Increase in sales in 2002-2004} = 632.5 - 500 = 132.5\text{M}$$

d) Compute the amount of external requirement of the firm over the 2 years of forecasting period.

$$\begin{aligned} \text{i) Increase in F. Assets} &= \% \text{ of sales} \times \text{increase in sales} \\ &= 60\% \times 132.5 = 79.5\text{M} \end{aligned}$$

$$\begin{aligned} \text{ii) Increase in C. Assets} &= \% \text{ of sales} \times \text{increase in sales} \\ &= 20\% \text{ of } 132.5 = 26.5\text{M} \end{aligned}$$

$$\text{Total additional investment/asset required} = \underline{106\text{M}}$$

Interpretation

For the company to earn increase in sales of 132.5M it will have to acquire additional assets costing 106M

	Sh.'000'
Additional investment/asset required	106,000
Less: Spontaneous source of finance	
Increase in creditors = % of sales x increase in sales	
= 132,500 x 10%	(13,250)
Increase in accrued expenses = % of sales x increase in sales	
= 132,500 x 6%	(7,950)
Less: Retained earnings during 2 years of operation (initial sources)	
Net profit for 2003 = Net profit margin x sales of 2003	
= 12% of 575,000	= 69,000
Less: Dividend payable 80% of 69,000	= <u>55,200</u> (13,800)

Net profit for 2004	= Net profit margin x sales of 2004	
	= 12% of 632,500	= 75,900
Less: dividend payable 80% of 75,900	=	<u>60,720</u> <u>(15,180)</u>
External financial needs (commercial paper)		<u>55,820</u>

Proforma Balance Sheet

This refers to the projected balance sheet at the end of forecasting period. The items in the proforma balance which vary with sales would be determined in any of the following two ways:

- i) % of sales x sales at last year of forecasting (2004); or
- ii) Balance sheet item before forecasting plus increase in balance sheet item as a result of increase in sales.

Proforma balance sheet as at 31st December 2004

	Shs.
Net fixed assets 60% x 632.5 or 300 + 79.5	379.50
Current Assets 20% x 632.5 or 100 + 26.5	<u>126.50</u>
	<u>506.00</u>
Ordinary shares (will remain constant)	100.00
Retained earning 70 + 13.8 + 15.18	98.98
10% debenture (remain constant)	150.00
Trade creditor 10% x 632.5 or 50 + 13.25	63.25
Accrued expenses 6% x 632.5 or 30 + 7.95	37.95
External borrowing - commercial	<u>55.82</u>
	<u>506.00</u>

TOPIC 3: THE TIME VALUE OF MONEY (TVM)

A dollar on hand today is worth more than a dollar to be received in the future because the dollar on hand today can be invested to earn interest to yield more than a dollar in the future. The Time Value of Money mathematics quantify the value of a dollar through time. This, of course, depends upon the rate of return or interest rate which can be earned on the investment.

The Time Value of Money has applications in many areas of Corporate Finance including Capital Budgeting, Bond Valuation, and Stock Valuation. For example, a bond typically pays interest periodically until maturity at which time the face value of the bond is also repaid. The value of the bond today, thus, depends upon what these future cash flows are worth in today's dollars.

The Time Value of Money concepts will be grouped into two areas: Future Value and Present Value. Future Value describes the process of finding what an investment today will grow to in the future. Present Value describes the process of determining what a cash flow to be received in the future is worth in today's dollars.

Concepts

- Future Value of a Single Cash Flow
- Present Value of a Single Cash Flow
- Cash Flow Streams
- Annuities
- Other Compounding Periods
- Equations

Future Value

The Future Value of a cash flow represents the amount, at some time in the future, that an investment made today will grow to if it is invested to earn a specific interest rate. For example, if you were to deposit Kshs 100 today in a bank account to earn an interest rate of 10% compounded annually, this investment will grow to Kshs 110 in one year. This can be shown as follows:

Year 1
$Kshs\ 100(1 + 0.10) = Kshs\ 110$

At the end of two years, the initial investment will have grown to Kshs 121. Notice that the investment earned Kshs 11 in interest during the second year, whereas, it only earned Kshs 10 in interest during the first year. Thus, in the second year, interest was earned not only on the initial investment of Kshs 100 but also on the Kshs 10 in interest that was paid at the end of the first year. This occurs because the interest rate in the example is a compound interest rate.

Compound Interest

Under compound interest, interest is earned not only on the initial principal but also on the accumulated interest. Interest begins to be earned on the accumulated interest as soon as it is paid, which occurs at the end of each compounding period. This is in contrast to **simple interest**, under which interest is only earned on the initial principal.

Valuations should generally be based on compound interest because, after the interest has been paid, the full amount, i.e., the initial principal plus interest, could be withdrawn and reinvested elsewhere. Thus, interest on the new investment would be earned on the full amount.

The interest rate in the example is 10% compounded annually. This implies that interest is paid annually. Thus the balance in the account was Kshs 110 at the end of the first year. Thus, in the second year the account pays 10% on the initial principal of Kshs 100 and the Kshs 10 of interest earned in the first year. Thus, the Kshs 121 balance in the account after two years can be computed as follows:

Year 2
Kshs $110(1+0.10) = \text{Kshs } 121$ or
Kshs $100(1+0.10)(1+0.10) = \text{Kshs } 121$ or
Kshs $100(1+0.10)^2 = \text{Kshs } 121$

If the money was left in the account for one more year, interest would be earned on Kshs 121, i.e., the initial principal of Kshs 100, the Kshs 10 in interest paid at the end of year 1, and the Kshs 11 in interest paid at the end of year 2. Thus the balance in the account at the end of year three is Kshs 133.10. This can be computed as follows:

Year 3
Kshs $121(1+0.10) = \text{Kshs } 133.10$ or
Kshs $100(1+0.10) (1+0.10) (1+0.10) = \text{Kshs } 133.10$ or
Kshs $100 (1+0.10)^3 = \text{Kshs } 133.10$

A pattern should be becoming apparent. The Future Value of an initial investment at a given interest rate compounded annually at any point in the future can be found using the following equation:

$$FV_t = CF_0(1+r)^t$$

where

- FV_t = the Future Value at the end of year t ,
- CF_0 = the initial investment,
- r = the annually compounded interest rate, and
- t = the number of years.

Future Value Example

Find the Future Value at the end of 4 years of Kshs 100 invested today at an interest rate 10%.

Solution:

$$FV_4 = 100(1 + .10)^4 = \$146.41$$

Example Problems

Find the Future Value of the following cash flow at the end of the given number of years.

Cash Flow: Kshs

When:

years

Interest Rate:

%

Future Value: Kshs

Present Value

Present Value describes the process of determining what a cash flow to be received in the future is worth in today's dollars. Therefore, the Present Value of a future cash flow represents the amount of money today which, if invested at a particular interest rate, will grow to the amount of the future cash flow at that time in the future. The process of finding present values is called *Discounting* and the interest rate used to calculate present values is called the *discount rate*. For example, the Present Value of Kshs 100 to be received one year from now is Kshs 90.91 if the discount rate is 10% compounded annually. This can be demonstrated as follows: (Refer to the Future Value page if you are unfamiliar with the calculations.)

One Year
$\text{Kshs } 90.91(1 + 0.10) = \text{Kshs } 100 \text{ or}$
$\text{Kshs } 90.91 = \text{Kshs } 100 / (1 + 0.10)$

Notice that the Future Value Equation is used to describe the relationship between the present value and the future value. Thus, the Present Value of Kshs 100 to be received in two years can be shown to be Kshs 82.64 if the discount rate is 10%.

Two Years
$\text{Kshs } 82.64(1 + 0.10)^2 = \text{Kshs } 100 \text{ or}$
$\text{Kshs } 82.64 = \text{Kshs } 100 / (1 + 0.10)^2$

A pattern should be becoming apparent. The following equation can be used to calculate the Present Value of a future cash flow given the discount rate and number of years in the future that the cash flow occurs. (This equation can be obtained algebraically from the Future Value Equation.)

$$PV = \frac{CF_t}{(1 + r)^t}$$

where

- PV = Present Value
- CF_t = Future Cash Flow which occurs t years from now
- r = the interest or discount rate
- t = the number of years

Present Value Example
Find the Present Value of Kshs 100 to be received 3 years from today if the interest rate is 10%

Solution:

$$PV = \frac{100}{(1 + .10)^3} = \$75.13$$

Example Problems

Find the Present Value of the following future cash flow.

Cash Flow: Kshs When: years

Discount Rate: % Present Value: Kshs

Cash Flow Streams

Present Value

The Present Value of a Cash Flow Stream is equal to the sum of the Present Values of the individual cash flows. To see this, consider an investment which promises to pay Kshs 100 one year from now and Kshs 200 two years from now. If an investor were given a choice of this investment or two alternative investments, one promising to pay Kshs 100 one year from now and the other promising to pay Kshs 200 two years from now, clearly, he would be indifferent between the two choices. (*Assuming that the investments were all of equal risk, i.e., the discount rate is the same.*) This is because the cash flows that the investor would receive at each point in time in the future are the same under either alternative. Thus, if the discount rate is 10%, the Present Value of the investment can be found as follows:

Present Value of the Investment

$$PV = \text{Kshs } 100/(1 + 0.10) + \text{Kshs } 200/(1 + 0.10)^2$$

$$PV = \text{Kshs } 90.91 + \text{Kshs } 165.29 = \text{Kshs } 256.20$$

The following equation can be used to find the Present Value of a Cash Flow Stream.

$$PV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$$

where

- PV = the Present Value of the Cash Flow Stream,

- CF_t = the cash flow which occurs at the end of year t ,
- r = the discount rate,
- t = the year, which ranges from zero to n , and
- n = the last year in which a cash flow occurs.

Present Value Example

Find the Present Value of the following cash flow stream given that the interest rate is 10%.

Year	0	1	2	3	4
Cash Flow		100	200	200	300

Solution:

$$PV = \frac{100}{(1+.10)^1} + \frac{200}{(1+.10)^2} + \frac{200}{(1+.10)^3} + \frac{300}{(1+.10)^4} = \$611.37$$

Future Value

The Future Value of a Cash Flow Stream is equal to the sum of the Future Values of the individual cash flows. For example, consider an investment which promises to pay Kshs 100 one year from now and Kshs 200 two years from now. Given that the discount rate is 10%, the Future Value at the end of year 2 of the investment can be found as follows:

Future Value of the Investment

$$FV_2 = \text{Kshs } 100(1 + 0.10) + \text{Kshs } 200$$

$$FV_2 = \text{Kshs } 110.00 + \text{Kshs } 200.00 = \text{Kshs } 310.00$$

As of year 2, the Kshs 100 received at the end of year 1 would have earned interest for one year while the Kshs 200 received at the end of year 2 would not yet have earned any interest. Thus, the Future Value at the end of year 2, *i.e.*, immediately after the Kshs 200 cash flow was received, is Kshs 310.00.

The following equation can be used to find the Future Value of a Cash Flow Stream at the end of year t .

$$FV_n = \sum_{t=0}^n CF_t (1+r)^{n-t}$$

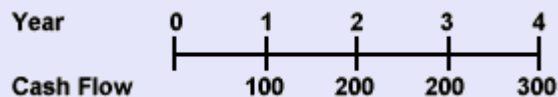
where

- FV_t = the Future Value of the Cash Flow Stream at the end of year t ,
- CF_t = the cash flow which occurs at the end of year t ,
- r = the discount rate,

- t = the year, which ranges from zero to n , and
- n = the last year in which a cash flow occurs.

Future Value Example

Find the Future Value at the end of year 4 of the following cash flow stream given that the interest rate is 10%.

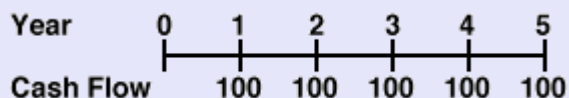


Solution:

Annuities

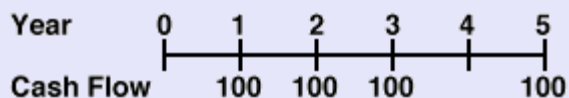
An Annuity is a cash flow stream which adheres to a specific pattern. Namely, an Annuity is a cash flow stream in which the cash flows are level (*i.e.*, all of the cash flows are equal) and the cash flows occur at a regular interval. The annuity cash flows are called *annuity payments* or simply *payments*. Thus, the following cash flow stream is an annuity.

Figure 1



While, the following cash flow stream is not an annuity because the payments do not occur at a regular interval.

Figure 2



When a cash flow stream is of the form given in **Figure 1**, *i.e.*, an annuity, the process of finding the Present Value or Future Value of the cash flow stream is greatly simplified.

Present Value of an Annuity

The Present Value of an Annuity is equal to the sum of the present values of the annuity payments. This can be found in one step through the use of the following equation:

$$PVA = PMT \left[\frac{1 - (1 + r)^{-t}}{r} \right]$$

where

- PVA = The Present Value of the Annuity
- PMT = The Annuity Payment
- r = The Interest or Discount Rate
- t = The Number of Years (also the Number of Annuity Payments)

Consider the annuity of Kshs 100 per year for five years given in **Figure 1**. If the discount rate is equal to 10%, then the Present Value of the Annuity can be found as follows:

Present Value of the Annuity
$PVA = 100 \left[\frac{1 - (1 + 0.10)^{-5}}{0.10} \right] = \379.08

Example Problems

Find the Present Value of the following Annuity.

Payment: Kshs Years:

Discount Rate: % Present Value: Kshs

Future Value of an Annuity

The Future Value of an Annuity is calculated at the end of the period in which the last annuity payment occurs. The Future Value of the Annuity is equal to the sum of the future values of the individual annuity payments at that time. Thus, the future value of a five year annuity is computed at the end of year five. This can be found in one step through the use of the following equation:

$$FVA_t = PMT \left[\frac{(1 + r)^t - 1}{r} \right]$$

where

- FVA = The Present Value of the Annuity
- PMT = The Annuity Payment
- r = The Interest or Discount Rate
- t = The Number of Years (also the Number of Annuity Payments)

Consider the annuity of Kshs 100 per year for five years given in **Figure 1**. If the discount rate is equal to 10%, then the Future Value of this Annuity at the end of period five can be found as follows:

Future Value of the Annuity

$$FVA_5 = 100 \left[\frac{(1 + 0.10)^5 - 1}{0.10} \right] = \$610.51$$

Example Problems

Find the Future Value of the following Annuity at the end of the given number of years.

Payment: Kshs Years:
Discount Rate: % Future Value: Kshs

Other Compounding Periods

In the real world, interest rates are often compounded more often than once per year. By convention, interest rates are quoted on an annual basis. An interest rate, quoted on an annual basis, which is compounded more often than once per year is called a *nominal rate, stated rate, quoted rate, or annual percentage rate (APR)*. For example, mortgages typically require monthly payments and, therefore, the interest rates quoted on mortgages are compounded monthly. Thus, the nominal interest rate on a mortgage might be 12% compounded monthly. However, the relevant rate for valuations is the periodic rate. The periodic rate is computed by dividing the nominal rate by the number of compounding periods per year.

$$r = \frac{r_{\text{nom}}}{m}$$

where

- r = the rate per period,
- r_{nom} = the nominal rate, and
- m = the number of compounding periods per year.

Thus a 12% nominal rate compounded monthly is equivalent to a periodic rate of 1% per month.

The following sections of this page demonstrate how to convert a nominal rate into an equivalent rate that is compounded annually and provide versions of the Present Value and Future Value formulas for use with interest rates compounded more often than once per year. The page concludes with a discussion of continuous compounding.

EAR - Effective or Equivalent Annual Rate

The Effective or Equivalent Annual Rate (EAR) is the interest rate compounded annually that is equivalent to a nominal rate compounded more than once per year. In other words, present and future values computed using the EAR will be the same as those computed using the nominal rate. The EAR is computed as follows:

$$EAR = \left(1 + \frac{r_{nom}}{m}\right)^m - 1$$

- EAR = the Equivalent or Effective Annual Rate,
- r_{nom} = the nominal interest rate,
- m = the number of compounding periods per year, and

Moreover, it is not proper to directly compare interest rates which have a particular compounding frequency with those that have a different compounding frequency, *e.g.*, comparing 10.1% compounded semiannually with 10% compounded quarterly. This problem can be overcome by finding the EAR for each of the rates and then comparing the EARs.

First, let's find the EAR for 10.1% compounded semiannually. Here, m equals 2.

EAR for 10.1% compounded semiannually
$EAR = \left(1 + \frac{0.101}{2}\right)^2 - 1 = 0.1036 = 10.36\%$

Now, let's find the EAR for 10% compounded quarterly. Here $m = 4$.

EAR for 10% compounded quarterly
$EAR = \left(1 + \frac{0.10}{4}\right)^4 - 1 = 0.1038 = 10.38\%$

Thus, we see that 10% compounded quarterly is actually a higher interest rate than 10.1% compounded semiannually. Given a choice, we would prefer to invest at 10% compounded quarterly.

Present Value

The Present Value of a future cash flow when the interest rate is compounded m times per year can be calculated as follows:

$$PV = \frac{CF_t}{\left(1 + \frac{r_{nom}}{m}\right)^{mt}}$$

where

- PV = the Present Value,
- CF_t = the cash flow which occurs at the end of year t,
- r_{nom} = the nominal interest rate,
- m = the number of compounding periods per year, and
- t = the number of years.
- Thus, mt = the number of compounding periods in t years.

In the earlier discussion of Present Value the interest rate was compounded annually and there was one compounding period per year. In that case $m = 1$. Thus, our earlier Present Value formula is actually just a special case of this formula since under annual compounding the rate per period is the same as the nominal rate.

Present Value Example

Find the Present Value of Kshs 100 to be received 3 years from today if the interest rate is 12% compounded quarterly.

Solution:

$$PV = \frac{100}{\left(1 + \frac{.12}{4}\right)^{4(3)}} = \$70.14$$

Future Value

The Future Value of a future cash flow when the interest rate is compounded m times per year can be calculated as follows:

$$FV_t = CF_0 \left(1 + \frac{r_{nom}}{m}\right)^{mt}$$

where

- FV = the Future Value,
- CF_0 = the cash flow which occurs at time 0,
- r_{nom} = the nominal interest rate,
- m = the number of compounding periods per year, and
- t = the number of years.
- Thus, mt = the number of compounding periods in t years.

Thus, the earlier Future Value formula is actually just a special case of this formula

since under annual compounding (*i.e.*, when $m = 1$) the rate per period is the same as the nominal rate.

Future Value Example

Find the Future Value of 3 years from now of Kshs 100 invested today at an interest rate of 10% compounded semiannually.

Solution:

$$FV_3 = 100 \left(1 + \frac{.10}{2} \right)^{2(3)} = \$134.01$$

Present Value of an Annuity

The Present Value of an Annuity when the payments occur m times per year and the interest rate is compounded m times per year can be calculated as follows:

$$PVA = PMT \left[\frac{1 - \left(1 + \frac{r_{nom}}{m} \right)^{-mt}}{\frac{r_{nom}}{m}} \right]$$

where

- PVA = the Present Value,
- PMT = the Annuity Payment which occurs m times per year,
- r_{nom} = the nominal interest rate,
- m = the number of compounding periods per year, and
- t = the number of years.
- Thus, mt = the number of payments and compounding periods in t years.

This formula can only be applied when the frequency of the annuity payments is the same as the compounding period for the interest rate. For example, if the annuity has quarterly payments the interest rate must be compounded quarterly ($m = 4$).

Thus, the earlier Present Value on an Annuity formula is actually just a special case of this formula since under annual compounding (*i.e.*, when $m = 1$) the rate per period is the same as the nominal rate.

Present Value of an Annuity Example

Find the Present Value of an annuity of Kshs 100 per month for 2 years if the interest rate is 12% compounded monthly

Solution:

$$PVA = 100 \left[\frac{1 - \left(1 + \frac{.12}{12}\right)^{-12(2)}}{\frac{.12}{12}} \right] = \$2124.34$$

Future Value of an Annuity

The Future Value of an Annuity when the payments occur m times per year and the interest rate is compounded m times per year can be calculated as follows:

$$FVA_t = PMT \left[\frac{\left(1 + \frac{r_{nom}}{m}\right)^{mt} - 1}{\frac{r_{nom}}{m}} \right]$$

where

- FVA_t = the Future Value of the annuity at the end of year t ,
- PMT = the Annuity Payment which occurs m times per year,
- r_{nom} = the nominal interest rate,
- m = the number of compounding periods per year, and
- t = the number of years.
- Thus, mt = the number of payments and compounding periods in t years.

This formula can only be applied when the frequency of the annuity payments is the same as the compounding period for the interest rate. For example, if the annuity has quarterly payments the interest rate must be compounded quarterly ($m = 4$). As with the earlier formula, the Future Value is computed at the end of the period in which the last annuity payment occurs.

Thus, the earlier Future Value on an Annuity formula is actually just a special case of this formula since under annual compounding (*i.e.*, when $m = 1$) the rate per period is the same as the nominal rate.

Future Value of an Annuity Example

Find the Future Value at the end of 3 years of an annuity of Kshs 100 per quarter for 3 years if the interest rate is 8% compounded quarterly.

Solution:

$$FVA_3 = 100 \left[\frac{\left(1 + \frac{.08}{4}\right)^{4(3)} - 1}{\frac{.08}{4}} \right] = \$1341.21$$

Time Value of Money Equations

Annual Compounding

Present Value: $PV = \frac{CF_t}{(1+r)^t}$

Future Value: $FV_t = CF_0(1+r)^t$

Present Value of a Cash Flow Stream: $PV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$

Future Value of a Cash Flow Stream: $FV_n = \sum_{t=0}^n CF_t(1+r)^{n-t}$

Present Value of an Annuity: $PVA = PMT \left[\frac{1 - (1+r)^{-t}}{r} \right]$

Future Value of an Annuity: $FVA_t = PMT \left[\frac{(1+r)^t - 1}{r} \right]$

Other Compounding Periods

Rate per Period: $r = \frac{r_{nom}}{m}$

Effective Annual Rate: $EAR = \left(1 + \frac{r_{nom}}{m}\right)^m - 1$

Present Value: $PV = \frac{CF_t}{\left(1 + \frac{r_{nom}}{m}\right)^{mt}}$

Future Value:

$$FV_t = CF_0 \left(1 + \frac{r_{nom}}{m} \right)^{mt}$$

Present Value of an Annuity:

$$PVA = PMT \left[\frac{1 - \left(1 + \frac{r_{nom}}{m} \right)^{-mt}}{\frac{r_{nom}}{m}} \right]$$

Future Value of an Annuity:

$$FVA_t = PMT \left[\frac{\left(1 + \frac{r_{nom}}{m} \right)^{mt} - 1}{\frac{r_{nom}}{m}} \right]$$

TOPIC 4: INVESTMENT/CAPITAL BUDGETING/CAPITAL EXPENDITURE DECISIONS

Introduction to investment analysis

Any company will invest finance for the sake of deriving a return which is useful for four main reasons:

1. To reward the shareholders or owners of the business for staking their money and by foregoing their current purchasing power for the sake of current and future return.
2. To reward creditors by paying them regular return in form of interest and repayment of their principal as and when it falls due.
3. To be able to retain part of their earnings for plough back purposes which facilitate not only the company's growth present and the future but also has the implication of increasing the size of the company in sales and in assets.
4. For the increase in share prices and thus the credibility of the company and its ability to raise further finance.
Such a return is necessary to keep the company's operations moving smoothly and thus allow the above objective to be achieved.

A financial manager with present investment policies will be concerned with how efficiently the company's funds are invested because it is from such investment that the company will survive. Investments are important because:

- i) They influence company's size
- ii) Influence growth
- iii) Influence company's risks

In addition, this investment decision making process also known as capital budgeting, involves the decision to invest the company's current funds in viable ventures whose returns will be realized for long term periods in future. Capital budgeting as financial planning is characterized by the following:

1. Decisions of this nature are long term i.e. extending beyond one year in which case they are also expected to generate returns of long term in nature.
2. Investment is usually heavy (heavy capital injection) and as such has to be properly planned.
3. These decisions are irreversible and any mistake may cause the company heavy losses.

Importance of Investment Decisions

- a) Such decisions are importance because they will influence the company's size (fixed assets, sales, and retained earnings).
- b) They increase the value of the company's shares and thus its credibility.
- c) The fact that they are irreversible means that they have to be made carefully to avoid any mistake which can lead to the failure of such investment.

- d) Due to heavy capital outlay, more attention is required to avoid loss of huge sums of money which in the extreme may lead to the closure of such a company. However, these decisions are influenced by:
- i) **Political factors** – Under conditions of political uncertainty, such decisions cannot be made as it will entail an element of risk of failure of such investment. Thus political certainty has to be analyzed before such decisions are made, such factors must be taken into account such that the company forecasts the inflows and outflows within given limitations such as the degree of competition, performance of economy, changing tastes etc. which influence ability to generate sufficient return from a venture which will pay not only interest but principal on such funds invested.
- ii) **Technological factors** – These influence the returns of the company because such technology will affect the company's ability to utilize its assets to the utmost ability in particular if such assets become obsolete and cannot generate good returns or the output of such machines may be low with time and may not meet planned expectations which in most cases will have an impact on inflows from a

Investment Projects Appraisal Criteria

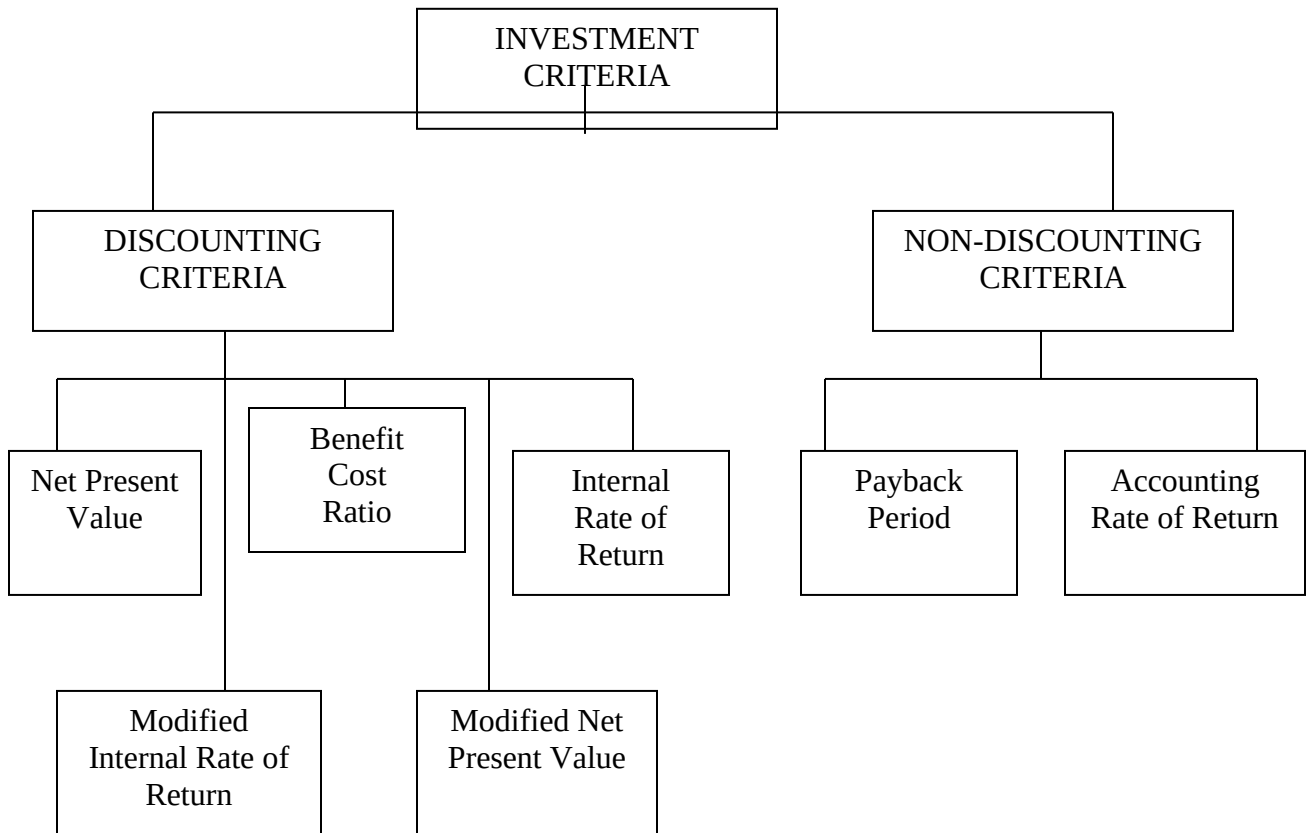
The key steps involved in determining whether a project is worthwhile or not are:-

- Estimate the costs and benefits of the project.
- Assess the riskiness of the project.
- Calculate the cost of capital.
- Compute criterion of merit and judge whether the project is viable or not.

There are several criteria that have been suggested by economists, accountants, financial analysts and others to judge the worthwhile ness of capital projects. These criteria are classified into two broad categories:

- (i) **Discounting criteria and**
- (ii) **Non-discounting criteria.**

Figure 1:
Discounted and Non-Discounted Appraisal
Techniques (Investment Criteria)



NON – DISCOUNTED CRITERIA /NON – DISCOUNTED CASH FLOW TECHNIQUES / TRADITIONAL METHODS

These methods don't explicitly consider the time value of money. In other words, each shilling/dollar earned in the future is assumed to have the same value as each shilling/dollar that was invested many years earlier. They include:

1. Accounting rate of return (ARR). Accounting rate of return, also referred to as the average return on investment is a measure of profitability which relates income to investment, both measured in accounting terms. Since income and investment can be measured in various ways, there can be a very large number of measures for accounting rate of return. The measures that are employed commonly in practice are:-

A: $\frac{\text{Average income after tax}}{\text{Initial investment}} \times 100$

B: $\frac{\text{Average income after tax}}{\text{Average investment}} \times 100$

C: $\frac{\text{Average income after tax but before interest}}{\text{Initial investment}} \times 100$

$$D: \frac{\text{Average income after tax but before interest}}{\text{Average investment}} \times 100$$

$$E: \frac{\text{Average income before interest \& taxes}}{\text{Initial investment}} \times 100$$

$$F: \frac{\text{Average income before interest and taxes}}{\text{Average investment}} \times 100$$

$$G: \frac{\text{Total income after tax but before depreciation} - \text{initial investment}}{(\text{Initial investment} / 2) \times \text{years}}$$

Illustration

Kisumu Gen Ltd intends to undertake a geothermal electricity generating project with a cost of Kshs.80 millions. The project is expected to have a life of 5 years and nil residual value. Expected earnings before depreciation and taxes are: Ksh.20 million, 24 million, 28 million, 32 million and 40 million during years 1 to 5 respectively. Corporate tax rate is 40% and depreciation is on a straight line method. Required: compute the projects accounting rate of return?

Solution

Step 1: Choose the method to adopt e.g. Method B:

$$ARR = \frac{\text{Average income after tax}}{\text{Average investment}} \times 100$$

Step 2: Compute average income after tax – use comprehensive statement of income (extract) as follows:

Kisumu Gen Ltd. Comprehensive statement of income (extract)

Years	1	2	3	4	5
	Shs.000	000	000	000	000
Earnings before depreciation & tax	20,000	24,000	28,000	32,000	40,000
Less: depreciation	(16,000)	(16,000)	(16,000)	(16,000)	(16,000)
Earnings before tax	4,000	8,000	12,000	16,000	24,000
Less tax (40%)	(1,600)	3,200	4,800	6,400	9,600
Earnings after tax	<u>2,400</u>	<u>4,800</u>	<u>7,200</u>	<u>9,600</u>	<u>14,400</u>

$$\text{Average income} = \frac{2400 + 4800 + 7200 + 9600 + 14400}{5}$$

$$\therefore \text{Average income} = \underline{7,680}$$

Step 3: compute the average investment

Use – statement of financial position (extract) as follows:-

Kisumu Gen Ltd
Statement of financial position

Years	1	2	3	4	5
	Shs.000	000	000	000	000
Book value of Investment - beginning	80,000	64,000	48,000	32,000	16,000
Less depreciation	<u>(16,000)</u>	<u>(16,000)</u>	<u>(16,000)</u>	<u>(16,000)</u>	<u>(16,000)</u>
Ending book value of investment	<u>64,000</u>	<u>48,000</u>	<u>32,000</u>	<u>16,000</u>	<u>0</u>

Average investment: Year 1 = $\frac{80,000 + 64,000}{2} = 72,000$

Year 2 = $\frac{64,000 + 48,000}{2} = 56,000$

Year 3 = $\frac{48,000 + 32,000}{2} = 40,000$

Year 4 = $\frac{32,000 + 16,000}{2} = 24,000$

Year 5 = $\frac{16,000 + 0}{2} = 8,000$

$\frac{200,000}{5} = 40,000$

Step 4: Substitute the values obtained in step 2 and 3 in the formula:

$ARR = \frac{\text{Average income after tax}}{\text{Average investment}} \times 100$

$= \frac{7,680}{40,000} \times 100 = 19.2\%$

Acceptance Criterion / Decision Rule

The management establishes the minimum accounting rate of return. Any project with an ARR higher than that established by the management (threshold level) would be accepted while any project with an ARR lower than the minimum rate that is fixed by the management would be rejected.

Ranking - The project with the highest ARR would be ranked as number one while the project with the lowest will come last.

Evaluation of ARR

Traditionally, a popular project appraisal criterion, the AAR has the following virtues:-

- It is simple to calculate.
- It is based on accounting information which is readily available and familiar to businessmen.
- It considers benefits over the entire life of the project.
- Since it is based on accounting measures, which can be readily obtained from the financial accounting, system of the firm, it facilitates post-auditing of capital expenditures. The short comings of the ARR are:-
 - It is based open accounting profit not cash flow
 - It doesn't take into account the time value of money
 - These are, as we have seen numerous measure of accounting rate of return. This can create controversy confusion and problems in interpretation.
- Accounting income (whatever particular measure of income we choose) is not uniquely defined because it is influenced by the methods of depreciation, inventory valuation, and allocation of certain costs.
- The argument test the accounting rate of return measures facilitate post-auditing of capital expenditure is not very valid. The financial accounting system of a form is designed to report events with respect to accounting periods and for profit centres but not for individual investment projects.
- The target or hurdle rate of return cannot be specified in relation to the goal of shareholder wealth maximization.
- The accounting rate of return method can be highly misleading because statement of financial position book values reflects neither the earning capacities of assets nor their marked values.

Payback Period

The payback period is the length of time required to recover the initial cash outlay on the projects. It is the number of years required to return the original investment from the net cash from (net operating income after taxes plus depreciation).

For example, if a project involves a cash outlay of Kshs.600,000 and generates cash inflows of shs100,000, shs.150,000, shs.150,000 and shs.200,000, in the first, second, third and four years respectively, it's payback period in 4 years because the sum of cash inflows during the 4 years is equal to the initial outlay. When the annual cash inflow is a constant sum, the payback period is simply the initial outlay divided by the annual cash inflow. For example, a project which has an initial cash outlay of shs.1, 000, 000 and a constant annual cash inflow of shs.300,000, has a payback period of $\text{shs.1,000,000} / 300,000 = 3 \frac{1}{2}$ years.

Acceptance Rule / Decision Criterion

According to payback criterion, the shorter the payback period, the more desirable the project is. Firms using this criterion generally specify the maximum acceptable payback period. If this is n years, project with payback period of n years or less is deemed worthwhile and projects with a payback period exceeding n years are considered unworthy.

Evaluation

A widely used investment criterion, the payback period seems to offer the following advantages.

- It is simple, both in concept and application.
- It is a rough and ready method for dealing with risk. It favors projects which generate substantial cash inflows in the earlier years, risk tend to increase with future.
- Since it emphasizes earlier cash flows, it may be a sensible criterion when the firm is pressed with problems of liquidity.

Limitations of the payback criterion include:

- It fails to consider the time value of money.
- It ignores cash flows beyond the payback period.
- It is a measure of the projects capital recovery, not profitability.

Discounted Cash Flow Techniques / Modern Methods

One on the best ways of determining if a project can make enough money to make it feasible is discounted cash flow (DCF) analysis. DCF analysis has many tools, the most useful are

- Net Present Values (NPV).
- Modified Net Present Value (MNPV).
- Internal Rate of Return.
- Modified Internal Rate of Return.
- Benefit – Cost Ratio (Profitability Index)

1. Net Present Value (NPV)

The net present value (NPV) of a project is the sum of the present values of all the cash flows positive as well as negative – that are expected to occur over the life of the project. The general formula for NPV is:

$$\text{NPV of a project} = \sum_{t=1}^n \frac{C_t}{(1+r)^t} - \text{Initial investment } (C_0)$$

where C_t = cash flow at the end of year t
 n = life of the project
 r = discount rate

The cash flows are the real cash flows from the project that are a result of the capital expenditure. Depreciation and loan repayments are added back to net profit to yield cash flow.

The discount rate to be used in the weighted average cost of capital for the investor of the current project belongs to the same risk – class with the already existing projects.

To illustrate the calculation of NPV, consider a project which has the following cash flow stream:

Year	Cash flow
0	Shs.(1,000,000)
1	200,000

2	200,000
6	300,000
4	300,000
5	350,000

The cost of capital, r for the firm is 10 percent.

The net present value of the proposed is

$$NPV = \frac{200,000}{(1.10)^1} + \frac{200,000}{(1.10)^2} + \frac{300,000}{(1.10)^3} + \frac{300,000}{(1.10)^4}$$

$$+ \frac{350,000}{(1.10)^4} - 1,000,000 = -\text{sh.}5,273$$

The net present value represents the net benefit over and above the compensation for time and risk.

Acceptance criterion / decision rule

Accept the project if the net present value is positive and reject the project if the net present value is negative.

Evaluation

A widely used investment criterion, the NPV offer the following advantages.

- It recognize the time value of money
- It uses cash flow and not profits and therefore it gives a reasonable assessment of the projects viability.
- It ranks projects according to their true profitability.
- It is consistent with the value additivity principle/consistent with shareholders wishes.

Limitations of NPV criterion include:-

- It is more complicated to use and to understand when compared to the traditional methods.
- It uses the firms cost of capital to evaluate the viability of investment project.
- It ignores risks associated with investment projects.
- It is very hard to determine the firm cost of capital.

Modified Net Present Value

The standard net present value method is based on the assumption that the intermediate cash flows are re invested at a rate of return equal to the cost of capital. When this assumption is not valid, the reinvestment rates applicable to the intermediate cash flows need to be defined for calculating the modified net present value. The steps involved in this process are as follows:

Step 1 – Calculate the terminal value of the projects cash inflows using the explicitly defined reinvestment rate(s) which is supposed to reflect the profitability of the investment opportunities ahead of the firm.

$$TV = \sum_{t=1}^n C F_t (1 + r)^{n-t}$$

Where TV = terminal value of the projects cash inflows.

$C F_t$ = Cash inflow at the end of year t

r = reinvestment rate applicable to the cash inflows of the project.

Step 2. Determine the modified net present value

$$NPV^* = \frac{TV}{(1 + r)^n} - I$$

Where NPV* - modified net present value

r = cost of capital

i = initial investment

n = period of investment

To illustrate the calculation of NPV* consider two projects X and Y:

	Project X	Project Y
Investment outlay	Shs.110,000	Shs.110,000
Cash inflows		
Year 1	31,000	71,000
Year 2	40,000	40,000
Year 3	50,000	40,000
Year 4	70,000	20,000

The NPV* for X and Y may be calculated for two reinvestment rates 14 percent and 20 percent, assuming a cost of capital of 10 percent.

Solution

Step 1. The terminal values of the cash inflows of project X and Y for the two reinvestments rates are:

$$TV(x)14\% = 31,000 (1.14)^3 + 40,000 (1.14)^2 + 50,000 (1.14)^1 + 70,000 = \text{Shs.}224,912$$

$$TV(x)20\% = 31,000 (1.20)^3 + 40,000 (1.20)^2 + 50,000 (1.20)^1 + 70,000 = \text{Shs.}241,168$$

$$TV(y)14\% = 71,000 (1.14)^3 + 40,000 (1.14)^2 + 40,000 (1.14)^1 + 20,000 = \text{shs.}223,911$$

$$TV(y)20\% = 71,000 (1.20)^3 + 40,000 (1.20)^2 + 40,000 (1.20)^1 + 20,000 = \text{shs.}249,488$$

Step 2: The NPV* of projects X and Y for the two reinvestment rates are

$$NPV^*(x)14\% = \frac{TV(x)14\%}{(1.10)^4} - 110,000 = \text{shs.}43,614$$

$$NPV^*(x)20\% = \frac{TV(x)20\%}{(1.10)^4} - 110,000 = \text{shs.}54,717$$

$$NPV^* (y) 14\% = \frac{TV (y) 14\%}{(1.10)^4} - 110,000 = \text{shs}42,614$$

$$NPV^* (x) 20\% = \frac{TV (y) 20\%}{(1.10)^4} - 110,000 = \text{shs}60,400$$

2. Benefit Cost Ratio (BCR)

There are two ways of defining the relationship between benefits and costs:

$$\text{Benefit cost ratio: } GBCR = \frac{PVB}{I}$$

$$\text{Net benefit cost ratio } NBCR = \frac{PVB - I}{I} = BCR - 1$$

Where PVB = present value of benefits

I = initial investment

To illustrate the calculation of these measures, let us consider a project which is being evaluated by a firm that has a cost of capital of 12 percent.

Initial investment	Shs.100,000
Benefits:	
Year 1	25,000
Year 2	40,000
Year 3	40,000
Year 4	50,000

Solution

The benefit cost ratio measures for this project are:-

$$BCR = \frac{\frac{25,000}{1.12} + \frac{40,000}{(1.12)^2} + \frac{40,000}{(1.12)^3} + \frac{50,000}{(1.12)^4}}{100,000} = 1.145$$

$$NBCR = BCR - 1 = 0.145$$

Acceptance criterion / decision rule

When GBCR or NBCR Rule is

> 1	> 0	Accept
= 1	= 0	Indifferent
< 1	< 0	Reject

Evaluation

The proponents of BCR argue that since this criterion measures net present value per shilling of outlay, it can discriminate better between large and small investment and hence preferable to the net present value criterion.

Internal Rate of Return (IRR)

The internal rate of return (IRR) of a project is the discount rate which makes its NPV equal to zero. It is the discount rate which equates the present value of future cash flows with the initial investment. It is the value of r in the following equation.

$$\text{Investment} = \sum_{t=1}^n \frac{C_t}{(1+r)^t}$$

Where C_t = cash flow at the end of year t

r = internal rate of return (IRR)

n = life of the project

In the NPV calculation we assume that the discount rate (cost of capital) is known and determine the NPV. In the IRR calculation, we set the NPV equal to zero and determine the discount rate that satisfies this condition.

$$\text{IRR estimate} = \frac{LR + \frac{LRPVIFA - PVIFA}{HRPVIFA - LRPVIFA} (HR - LR)}{1}$$

Where LR = lower rate

$LRPVIFA$ = lower rate present value interest factor of an annuity

$PVIFA$ = present value interest factor of an annuity

HR = Higher rate

$HRPVIFA$ = Higher rate present value interest factor of an annuity

Example

A project has a cost of Kshs.1 million and it is expected to generate cash flows of shs.360,000 each year for 4 years. The firm's cost of capital is 16%.

Required: calculate the project IRR

Solution

Present value of cash inflows = $360,000 \times PVIFA_{4 \text{ yrs } r \%}$

$$1,000,000 = 360,000 \times PVIFA_{4 \text{ yrs } r \%}$$

$$PVIFA_{4 \text{ yrs }, r \%} = \frac{1,000,000}{360,000} = 2.7773$$

IRR lies between 16% and 17% at 16% $PVIFA = 2.7982$
17% $PVIFA = 2.7402$

$$\begin{aligned} 1\% &= 16\% + \frac{0.0204}{0.055} \times 1 \\ &= 16.37\% \end{aligned}$$

Internal Rate of Return for a project that generates unequal or irregular cash flows

If a project generates irregular or unequal cash inflows, its IRR is determined by trial and error method and then interpolation.

The procedure is as follow: using trial and error method obtain two rates which are as close as possible one rate yielding negative NPV and the other **positive** NPV. Then interpolate them together with their respective NPV profits using the following formula:

$$\text{IRR estimate} = \text{LR} + \frac{+\text{NPV}}{+\text{NPV} - (-\text{NPV})} (\text{HR} - \text{LR})$$

Where LR = lower rate

NPV = positive NPV

-NPV = negative NPV

HR = higher rate

Example

A project has a cost of Kshs.324, 000 and is expected to have a useful life of 3 years. The cash inflows expected are shs.160, 000, Kshs.140, 000, and Kshs.120,000 at the end of years 1, 2, and 3 respectively.

Required: Calculate the projects IRR

Solution

Try 18 %

Year	Cash inflow	PVIF @18%	PV
1	160,000	0.8473	135,600
2	140,000	0.7182	100,548
3	120,000	0.6085	<u>72,020</u>
PV of cash inflows			309,168
Less cash outlay			<u>(320,000)</u>
NPV			- <u>14,832</u>

At 14 %

Year	Cash inflows	PVIF @14%	PV
1	160,000	0.8772	140,312
2	140,000	0.7895	107,730
3	120,000	0.6950	21,000
PV of cash inflows			32,8082
Less cash outflows			<u>(324,000)</u>
NPV			<u>+5082</u>

IRR that equates cash outflows to cash inflows is between 14% and 18% using interpolation:

$$14\% + \frac{5082}{19,914} \times 4 = 15.02 \%$$

Acceptance criterion / decision rule. The IRR of the project is compared with the firms cost of capital. If the projects IRR is higher than the firm's cost of capital, the project is accepted otherwise reject.

Evaluation

Advantages

- Is taken into account time value of money.
- It considers cash flows occurring over the entire life of the project to determine the projects viability.
- It ranks projects according to their true profitability.
- It uses cash flows not profits hence it gives a reasonable assessment of the projects viability.
- Calculation of the cost of capital is not a pre-condition for the rise of this method.

Disadvantages

- It may involve tedious calculation.
- It may yield multiple and negative rates which may not have any meaning.
- Not consistent with value additively principle.

Modified Internal Rate of Return (MIRR)

Despite NPV'S conceptual superiority, managers seem to prefer IRR over NPV because IRR – intuitively more appealing as it is a percentage measure. However IRR has its shortcoming. To overcome them use MIRR.

The procedure for calculating MIRR is as follows:

Step 1: Calculate the present value of the costs (PVC) associated with the project, using the cost of capital (r) as the discount rate.

$$PVC = \sum_{t=0}^n \frac{\text{Cash outflow } t}{(1+r)^t}$$

Step 2: Calculate the terminal value (TV) of the cash inflows expected from the project.

$$TV = \sum_{t=0}^n \text{cash inflows } t (1+r)^{n-t}$$

Step 3 – obtain MIRR by solving the following equation

$$PVC = \frac{TV}{(1 + \text{MIRR})^n}$$

To illustrate the calculation of MIRR let us consider an example. Pentagon Ltd in evaluating a project that has the following cash flow stream associated with it.

Year	0	1	2	3	4	5	6
Cash flow (Kshs in millions)	-120	-80	20	60	80	100	120

The cost of capital for pentagon is 15%

Solution

The present value of costs is

$$120 + \frac{80}{(1.15)} = 189.6$$

The terminal value of cash inflows is :

$$\begin{aligned} &= 20(1.15)^4 + 60(1.15)^3 + 80(1.15)^2 + 100(1.15) + 120 \\ &= 34.98 + 91.26 + 105.76 + 115 + 120 = 467 \end{aligned}$$

The MIRR is obtained as follows:

$$189.6 = \frac{467}{(1 + \text{MIRR})^6}$$

$$(1 + \text{MIRR})^6 = 2.463$$

$$1 + \text{MIRR} = 2.463^{1/6} = 1.162$$

$$\text{MIRR} = 1.162 - 1 = 0.162 \text{ or } 16.2\%$$

PRACTICE QUESTIONS

Attempt the following questions and compare your answers with the solution provided

QUESTION ONE

Assume a project costs Sh.80, 000 and will generate the following cash inflows:

Cash inflows	Accumulated inflows
Inflows year 1 = 10,000	10,000
Inflows year 2 = 30,000	40,000
Inflows year 3 = 15,000	55,000
Inflows year 4 = 20,000	75,000
Inflows year 5 = 30,000	105,000

Required: Calculate Payback Period

Solution

The Sh.80, 000 cost is recovered between year 4 and 5. During year 5 (after year 4) Sh.5,000 is (80,000 – 75,000) is required out the total year 5 cash flows of 30,000.

$$\text{Therefore the PBP} = 4\text{yrs} + \frac{5,000}{30,000} = 4.17 \text{ years}$$

QUESTION TWO

Cedes limited has the following details of two of the future production plans. Only one of these machines will be purchased and the venture would be taken to be virtually exclusive. The Standard model costs Kshs50,000 and the Deluxe cost Kshs88,000 payable immediately. Both machines will require the input of the following:

- i) Installation costs of Kshs 20,000 for Standard and Kshs 40,000 for the Deluxe
- ii) A Kshs10, 000, working capital through their working lives.

Both machines have no expected scrap value at end of their expected working lives of 4 years for the Standard machine and six years for the Deluxe. The operating pre-tax net cash flows associated with the two machines are:

Year	1	2	3	4	5	6
Standard	28,500	25,860	24,210	23,410	-	-
Deluxe	36,030	30,110	28,380	25,940	38,500	35,100

The deluxe machine has only been introduced in the market and has not been fully tested in the operating conditions, because of the high risk involved the appropriate discount rate for the deluxe machine is believed to be 14% per annum, 2% higher than the rate of the standard machine. The company is proposing the purchase of either machine with a term loan at a fixed rate of interest of 11% per annum, taxation at 30% is

payable on operating cash-flows one year in arrears and capital allowance are available at 25% per annum on a reducing balance basis.

Required

For both the Standard and the Deluxe machines, calculate the payback period.

Solution

Establish the cash flows as follows:

Pre-tax inflows (EBDT)		XX
Less depreciation = capital allowance		<u>(XX)</u>
Earnings before tax	XX	
Less tax		<u>(XX)</u>
Earnings after tax	XX	
Add back capital allowance/depreciation		<u>XX</u>
Operating cash flows	<u>XX</u>	

Note

Capital allowance/depreciation is a non-cash item thus when deducted for tax purposes, it should be added back to eliminate the non-cash flow effects.

Cash flows for standard machine:

Year	1	2	3	4	5
Pretax inflow	28,500	25,850	24,210	23,410	-
Less allowance (depreciation)	<u>17,500</u>	<u>13,125</u>	<u>9,844</u>	<u>7,383</u>	-
Taxable cash inflows	11,000	12,735	14,366	16,027	
Tax @ 30% 1 yr in arrears	<u>-</u>	<u>3,300</u>	<u>(3,831)</u>	<u>(4,310)</u>	<u>(4,808)</u>
	11,000	9,435	10,545	11,717	<u>(4,808)</u>
Add back capital allowance	<u>17,500</u>	<u>13,125</u>	<u>9,844</u>	<u>7,383</u>	-
Operating cash flows	28,500	22,560	20,389	19,100	(4,808)
Add working capital realised	<u>-</u>	<u>-</u>	<u>-</u>	<u>10,000</u>	<u>-</u>
Total cash flows	<u>28,500</u>	<u>22,560</u>	<u>20,389</u>	<u>29,100</u>	<u>(4,808)</u>

Cash flows for Deluxe machine

Year	1	2	3	4	5	6	7
Pretax inflows	36,030	30,110	28,380	25,940	38,560	35,100	-
Less	<u>32,000</u>	<u>24,000</u>	<u>18,000</u>	<u>13,500</u>	<u>10,125</u>	<u>7,594</u>	-
(depreciation)	4,030	6,110	10,380	12,440	28,435	27,506	-
	<u>-</u>	<u>(1,209)</u>	<u>(1,833)</u>	<u>(3,114)</u>	<u>(3,732)</u>	<u>(8,531)</u>	<u>(8,252)</u>
Tax @ 30% in arrears	4,030	4,901	8,547	9,326	24,703	18,975	(8,252)

Inflows after tax	<u>32,000</u>	<u>24,000</u>	<u>18,000</u>	<u>13,500</u>	<u>10,125</u>	<u>7,594</u>	<u>-</u>
Add back capital	-	-	-	-	-	10,000	-
Allowance	<u>36,030</u>	<u>28,599</u>	<u>26,547</u>	<u>22,826</u>	<u>34,828</u>	<u>36,569</u>	<u>(8,252)</u>

Add back
w/capital
Total cash
flows

		Standard		Deluxe
Cost	50,000 + 20,000	<u>70,000</u>		88,000 + 40,000
	<u>128,000</u>			

Year	Cash flows	Accumulated	Cash flows	Accumulated
1	28,500	28,500	36,030	36,030
2	22,560	51,060	28,901	64,931
3	20,389	71,449	26,547	91,478
4	29,100	100,549	22,826	114,304
5	(4,808)	95,741	34,828	149,132
6	-	-	36,569	185,701
7	-	-	(8,252)	179,449

* Pay back period for standard: Initial capital of Sh.7,000 is recovered during year 3. After year 2, we require 70,000 – 9,060 = 18,940 to recover initial capital out of year 3 cash flows of Sh.20,389.

* Applying the same concept for Deluxe, payback period would be:

$$4 + \frac{128,000 - 114,304}{34,828} = 4.39 \text{ years}$$

QUESTION THREE

Consider a project with the following characteristics:

Initial cost	Kshs 500,000
Scrap value	Kshs 100,000

Stream of income before depreciation and taxes are as follows:

	Kshs.
Year 1	100,000
Year 2	120,000
Year 3	140,000
Year 4	160,000
Year 5	200,000

Let tax = 50% and depreciation straight line.

Required: Calculate the accounting rate of return

Solution

$$\text{Depreciation} = \frac{500,000 - 100,000}{5 \text{ years}} = \text{Shs.80,000}$$

Year	1	2	3	4	5
Income	100,000	120,000	140,000	160,000	200,000
Less depreciation	<u>80,000</u>	<u>80,000</u>	<u>80,000</u>	<u>80,000</u>	<u>80,000</u>
Earnings before tax	20,000	40,000	50,000	80,000	120,000
EBT	<u>(10,000)</u>	<u>(20,000)</u>	<u>(30,000)</u>	<u>(40,000)</u>	<u>(60,000)</u>
Less tax @ 50%	<u>10,000</u>	<u>20,000</u>	<u>30,000</u>	<u>40,000</u>	<u>60,000</u>
EAT					

$$\text{Average income (EAT)} = 32,000$$

$$\text{Average investment} = (500,000 + 100,000) \frac{1}{2} = 300,000$$

$$\text{Or ARR} = \frac{\text{Average income}}{\text{Average investment}} \times 100 = \frac{32,000}{300,000} \times 100 = \underline{10.67\%}$$

Note

The best method of depreciation to use should be that which will produce larger depreciation charges in the 1st few years of the assets life and lesser changes in the later years because this will produce a higher tax shield to the company with higher value of inflows. Thus reducing balance is preferred as compared to sum of digits and straight line method.

The salvage value should be treated as follows:

If the asset produces a salvage value at the end of the year, this will increase inflows for payback period. This value is only used to ascertain how much the company will reduce original cost of investment to obtain average investment.

QUESTION FOUR

A project costs Kshs 16,200/= and is expected to generate the following inflows:

	Kshs
Year 1	8,000
Year 2	7,000
Year 3	6,000

Compute the IRR of this venture.

Solution

1st choice 10%

$$\frac{8,000}{(1.1)^1} + \frac{7,000}{(1.1)^2} + \frac{6,000}{(1.1)^3} = 17,565.74 > \text{cost, choose a higher rate.}$$

2nd choice 14%

$$\frac{8,000}{(1.14)^1} + \frac{7,000}{(1.14)^2} + \frac{6,000}{(1.14)^3} = 16,453.646$$

3rd choice 15%

$$\frac{8,000}{(1.15)^1} + \frac{7,000}{(1.15)^2} + \frac{6,000}{(1.15)^3} = 16,194.625$$

IRR lies between 14% and 15%.

ii) Interpolation method

PV at rate of 14%	=	16,453.646	Difference
PV required	=	16,200.000	253.646
PV at rate of 15%	=	16,194.625	-5.375

Therefore, r denotes required rate of return

$$\begin{aligned} \text{Therefore, } r &= 14\% + (15\% - 14\%) \times \frac{253.646}{253.646 + 5.375} \\ &= 14\% + 0.98\% \\ &= \underline{\underline{14.98\%}} \end{aligned}$$

QUESTION FIVE

The following information was from XYZ feasibility studies. It has studied two ventures:

- Cost 100,000/= and 160,000/= at the beginning of the 4th year and it will generate inflows 1-3rd year 80,000/= and from 4-6th year 50,000/= per annum.
- Initial cost 200,000/= and 80,000/= at the beginning of the 4th year and it will generate the following inflows:

1st – 2nd year -> Shs.100,000 per annum

3rd – 6th year -> Shs.70,000 per annum

Using the cost of finance of 12% compute the P.I. of these two ventures, advise the company accordingly.

Solution

$$\text{a) Outflows: } \frac{100,000}{1} + \frac{160,000}{(1.12)^3} = 100,000 + 113,887 = 213,885$$

$$\text{Inflows: } \frac{80,000}{(1.12)^1} + \frac{80,000}{(1.12)^2} + \frac{80,000}{(1.12)^3} + \frac{50,000}{(1.12)^4} + \frac{50,000}{(1.12)^5} + \frac{50,000}{(1.12)^6} = \text{Shs.}277,626$$

$$P.I. = \frac{277,626}{213,885}$$

$$P.I. = 1.298$$

$$b) \text{ Outflows: } = \frac{200,000}{1} + \frac{80,000}{(1.12)^3} = 256,944$$

$$\text{Inflows} = \frac{100,000}{(1.12)^1} + \frac{100,000}{(1.12)^2} + \frac{70,000}{(1.12)^3} + \frac{70,000}{(1.12)^4} + \frac{70,000}{(1.12)^5} + \frac{70,000}{(1.12)^6} = \text{Shs.}338,501$$

$$P.I. = \frac{338,501}{256,944}$$

$$= 1.32$$

QUESTION SIX

A company is faced with the following 5 investment opportunities:

	Cost	NPV	$P.I. = \frac{\text{Total P.v}}{\text{Initial capital}}$	P.I Ranking
1.	500,000	150,000	1.3	4
2.	100,000	40,000	1.4	3
3.	400,000	40,000	1.1	5
4.	200,000	100,000	1.5	2
5.	160,000	90,000	1.6	1

This company has 750,000/= available for investment projects, 3 and 4 are mutually exclusive. All of the projects are divisible. Which group should be selected in order to maximise the NPV. Indicate this NPV figure.

Solution: Using P.I. to rank the projects in order of preference 5, 4, 2, 1, 3.

In order to maximise NPV, the following projects combination should be selected:

		Shs.
Funds available for investment		750,000
Cost of project:	5	160,000
	4	200,000
	2	100,000
	1	<u>290,000</u>
		<u>(750,000)</u>
		<u>NIL</u>

$$NPV = 90,000 + 100,000 + 40,000 + \frac{290,000}{500,000} \times 150,000 = 317,000$$

TOPIC 5: MEASUREMENT OF BUSINESS PERFORMANCE/FINANCIAL RATIO ANALYSIS/FINANCIAL STATEMENT ANALYSIS

Definition

Financial ratio analysis looks at various ratios calculated from data taken from the financial statements. Financial ratios should be examined in the context of:
The firm's own financial history, known as trend analysis, and the industry in which the firm operates, known as cross-section analysis.

It is a process by which finance identifies the company's financial performances by comparing the entities in the statement of financial position and those in the income statement. This is so because statement of financial position commonly known as balance sheet entities are usually responsible for those to be found in the income statement i.e. assets shown in the balance sheet are responsible for sales, revenue and expenses to be found in the income statement. This analysis is important to various parties with a financial stake in the company. These include:

1. **Shareholders** – Actual owners are interested in the company's both short and long term survival. For this reason they will use ratios such as:
 - a) Profitability ratios – which seek to establish viability.
 - b) Dividend ratios – which seek to establish return to owners in form of dividends. The common ratios include earning yield (E/Y), Dividend pay- out ratio (DPO), dividend yield, Price- earnings ratio, all of which will measure return to owner.
2. **Creditors (trade)** – these are interested in the company's ability to meet their short-term obligations as and when they fall due. For this reason they will use ratios such as:
 - a) Liquidity ratio – a qualitative measure of company's liquidity position measured by acid test ratio.
 - b) Current ratio – which is a measure of company's quantity of current assets against current liabilities.
3. **Long term lenders** – These include finances through loans, mortgages and debenture holders. These have both short and long term interest in the company and its ability to pay not only interest on debt but also principal as and when it falls due. These parties are interested in the following:
 - a) Liquidity ratios – used to assess short-term liability to meet current obligations.
 - b) Profitability ratios – used to ascertain whether the company can pay its principal back.
 - c) Gearing ratio – used to gauge the company's risk in the investment.
 - d) Investment coverage ratio – shows the company's safety as regards the payment of interest to the lenders of the debt.
4. **Directors and management of company** – They will therefore be interest in:

- a) Efficiency of the company in generating profits.
 - b) The company's viability from the investor's point of view and the company's ability to generate sufficient returns to investors.
 - c) Gearing ratio to gauge the safety and risk associated with the company.
5. **Potential investors** – these parties are interested in a company in total both on short and long term basis in particular the company's ability to generate acceptable return on their money.

Therefore, they will use:

- a) Dividend ratios
 - b) Return ratios
 - c) Gearing ratios
6. **Government** – The Government is interested mostly in utility companies (e.g. KPLC, Telkom) and those that will provide public services – in this case the government will be interested in their survival and thus ability to provide those services. It may be interested in taxation derived from these companies which is used for development. Government may also be interested in employment level and as such it will use those ratios that can enable it to achieve such objectives of particular importance are:
- a) Profitability ratios
 - b) Return ratios
7. **Competitors** – These are interested in the company's performance from the market share point of view and will use the ratios that enable them to ascertain company's competitive strength e.g. profitability ratios, sales and returns ratio etc.
8. **General public** – Customers and potential customers – These are interested in the ability of the company to provide good services both in the short and long run. To gauge the company's ability to provide goods and services on short and long term basis. We have:
- a) Returns ratio
 - b) Sales ratio

BENCHMARKS/YARD STICKS USED IN RATIO ANALYSIS

1. Past performance of the company

The company's past performance (past ratio) is used to measure or gauge the company's performance and in particular the change in performance whether good (favorable), better, same or even worse than the past. Such comparison is then used to interpret the company's performance bearing in mind the factors that influenced the present and past performances.

2. Average industry ratios

These are useful as they indicate the average performance of various companies in a given industry i.e. it gives the minimum performance of a number of companies in a given industry. These ratios are useful in so far as to enable the analyst to make a reasonable comparison of the company's performance vis-à-vis other companies in the same industry. However, for this yardstick to be useful the term average should include

those companies which are not extremely. i.e. very strong and very weak companies – which should be excluded to arrive at industry average figures.

3. Ratio of successful companies

Useful if the company can get figures of competitors who are leading in the market so as to enable it to gauge its performance against better performance. However this information is difficult to obtain and sometimes it calls for private investigators e.g. Private Eyes Ltd.

4. Ratio of budgeted performance

These are compared with actual performance ratios and investigations are made of any unfavorable variance which should be explained.

CLASSIFICATION OF RATIOS

Ratios are broadly classified into 5 categories:

1. Liquidity ratios
2. Turnover ratios
3. Gearing ratios
4. Profitability ratios
5. Growth and valuation ratios

1. Liquidity Ratios

Also, called working capital ratios. They indicate ability of the firm to meet its short term maturing financial obligation/current liabilities as and when they fall due.

The ratios are concerned with current assets and current liabilities. They include:

a) Current ratio =
$$\frac{\text{Current Assets}}{\text{Current liabilities}}$$

This ratio indicates the No. of times the current liabilities can be paid from current assets before these assets are exhausted.

The most recommended ratio is 2.0 i.e. the current asset must at least be twice as high as current liabilities

b) Quick/acid test ratios =
$$\frac{\text{Current Asset} - \text{Stock}}{\text{Current liabilities}}$$

Is a more refined current ratio which exclude amount of stock of the firm. Stocks are excluded for two basic reasons.

- i) They are valued on historical cost basis
- ii) They may not be converted into cash very quickly

The ratio therefore indicates the ability of the firm to pay its current liabilities from the more liquid assets of the firm.

c) Cash ratio =
$$\frac{\text{Cash in hand/bank} + \text{short term marketable securities}}{\text{Current liabilities}}$$

This is a refinement of the acid test ratio indicating the ability of the firm to meet its current liabilities from its most liquid resources.

Short term marketable securities refers to short term investment of the firm which can be converted into cash within a very short period e.g commercial paper and treasury bills.

$$d) \quad \text{Net working capital Ratio} = \frac{\text{Networking Capital}}{\text{Net Assets}} \times 100$$

Where Net Assets or Capital employed = Total Assets – Current liability

This ratio indicates the proportions of total net assets which is liquid enough to meet the current liabilities of the firm.

It is expressed in % term.

2. Turnover Ratios/efficiency/asset management ratio

Turnover ratio indicates the efficiency with which the firm utilized the asset or resources at its disposal to generate sales revenue or turnover.

This ratio includes:

$$a) \quad \text{Stock/inventory turnover} = \frac{\text{Cost of Sales}}{\text{Average stock}}$$

The ratio indicates number of times the stock was turned into sales in a year i.e how many times did the 'buy-sell' process occur during the year. The higher the stock turnover, the better the firm and more likely, the higher the sales.

$$\begin{aligned} b) \quad \text{Stock holding period} &= \frac{365 \text{ days}}{\text{Stock turnover}} \\ &= \frac{365 \times \text{Average stock}}{\text{Cost of sales}} \quad \text{i.e.} \quad \frac{365}{\text{Stock turnover}} \end{aligned}$$

The ratio indicates number of days the stock was held in the warehouse before being sold.

The higher the stock turnover, the lower the stock holding period and vice versa.

$$c) \quad \text{Debtors/accounts receiver turnover} = \frac{\text{Annual credit sales}}{\text{Average debtors}}$$

The ratio indicate the number of times/frequency with which credit customers or debtors were turned into sale i.e the number of times they come to buy on credit per year after paying their dues to the firm.

The higher the debtors turnover the better the firm indicating that customers came to buy on credit many times thus they paid within a short period.

$$d) \quad \text{Debtors collection period} = \frac{365}{\text{Debtors turnover}}$$

$$\text{or } \frac{365 \times \text{Average debtors}}{\text{Annual credit sales}}$$

This refers to credit period that was granted to the debtors on the period within which they were supposed to pay their dues to the firm.

The shorter the collection period/credit period the higher the debtors turnover and vice versa.

If no opening debtors are given use the closing debtors to represent average debtors.

$$\text{e) Creditors/accounts payable turnover} = \frac{\text{Annual credit purchases}}{\text{Average creditors}}$$

Firms buy goods on credit from suppliers. The ratio indicates number of times p.a. the firm bought goods on credit after paying the suppliers.

If the creditors' turnover is high, this indicates that the payment was made within a short period of time.

$$\begin{aligned} \text{f) Creditors payment period} &= \frac{365}{\text{Creditors turnover}} \\ &= \frac{365 \times \text{Average creditors}}{\text{Annual credit purchases}} \end{aligned}$$

The ratio indicate the credit period granted by the suppliers i.e. the period within which the firm should pay its liabilities to the suppliers.

The shorter the period the higher the creditors turnover and vice-versa.

$$\text{g) Fixed asset turnover} = \frac{\text{Annual Sales}}{\text{Fixed Assets}}$$

This ratio indicate the efficiency with which, the fixed assets were utilised to generate sales revenue e.g. a ratio of 1.4 means one shilling of fixed assets was utilised to generate Sh.1.4 of sales.

$$\text{h) Total asset turnover} = \frac{\text{Annual sales}}{\text{Total assets}}$$

The ratio indicate amount of sales revenue generated from utilization of one shilling of total asset.

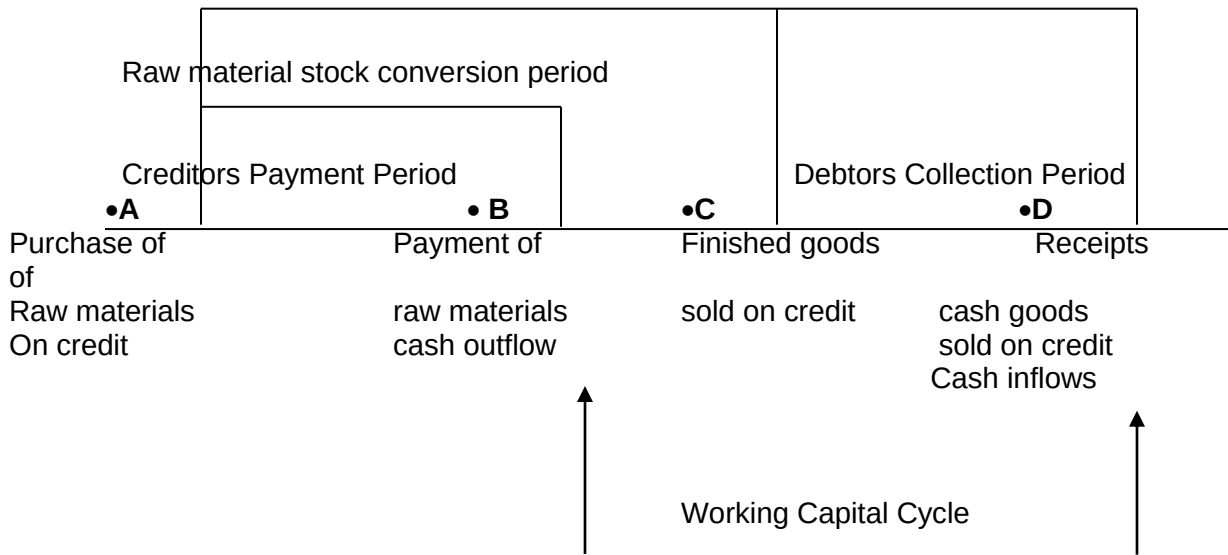
The Concept of Working Capital/Cash Operating Cycle

Working capital cycle refers to period that elapses between the payment for raw materials bought on credit (cash outflows) and the receipts of cash from finished goods sold on credit (cash inflows).

The working capital cycle will involve the following:

- a) Purchase of raw materials on credit from suppliers.
- b) Payment of raw materials after the lapse of credit period.
- c) Conversion of raw materials into finished goods.
- d) Sale of finished goods to creditors
- e) Receipt of cash from debtors.

This can be illustrated using a diagram as follows:



Working capital cycle = Stock conversion + debtors collection – Creditors payment

From the diagram the working capital cycle of a period will be determined as follows:

Stock conversion period + Debtors collection period – Creditors payment period

Note

A lengthy working capital cycle is an indicator of poor management of stock and debtors reflecting low turnover of stock and debtors and lengthy stockholding period and debtors collection period.

The working capital cycle can be reduced in any of the following ways:

1. Negotiate for a longer credit period with the suppliers
2. Reduce the stock conversion period or manufacturing period.
3. Reduce the debtor's collection period by granting short crediting period. This can be achieved through offering discounts to customers to encourage them to pay earlier.
4. Holding fast moving goods to ensure high turnover.
5. Timely delivery of raw materials by suppliers especially if any delay in delivery will lengthen the raw materials holding period.

3. Gearing/Leverage/Capital Structure Ratio

These ratios indicate the extent in which the firm has borrowed fixed charge capital to finance the acquisition of the assets or resources of the firm.

The two basic gearing ratios are:

$$\text{a) Debt/equity ratio} = \frac{\text{Fixed charge capital}}{\text{Equity (net worth)}}$$

This ratio indicates the amount of fixed charge capital in the capital structure of the firm for every one shilling of owners capital or equity e.g a ratio of 0.78 means for every Sh.1 of equity there is Sh.0.78 fixed charge capital.

$$\text{b) Fixed charge to total capital ratio} = \frac{\text{Fixed charge capital}}{\text{Total capital employed}} \times 100$$

Where total capital employed = Fixed charge capital + equity relative to total capital employed by the firm e.g a ratio of 0.38 means that, 38% of the capital employed is fixed charge capital.

Other leverage or gearing ratios are:

$$\text{a) Debt ratio} = \frac{\text{Total debts}}{\text{Total assets}}$$

Where total debt = fixed charge capital + liabilities.

The ratio indicates the proportion of total assets that has been financed using long term and current liabilities e.g a debt ratio of 0.45 means 45% of total asset has been financed with debt while the remaining 55% was financed with owners equity/capital.

$$\text{b) Times interest earned ratio} = \frac{\text{Operating profit (earnings before interest and tax)}}{\text{Interest Charges}}$$

TIER also called interest coverage ratio.

This ratio indicates the number of times interest charges can be paid from operating profit. The higher the TIER, the better the firm indicating that either the firm has high operating profits or its interest charges are low.

If TIER is high due to low interest charges, this indicates low level of gearing/debt capital of the firm.

4. Profitability Ratio

This ratio indicates the performance of the firm in relation to its ability to derive returns or profit from investment or from sale of goods i.e profit margin or sales.

1. Profitability in relation to sales

The ratio indicate the ability of the firm to control its cost of sales, operating and financing expenses.

They include:

$$\text{a) Gross profit margin} = \frac{\text{Gross profit} \times 100}{\text{Sales}}$$

The ratio indicate the ability of the firm to control cost of sales expenses e.g gross profit margin of 40% means 60% of sales revenue was taken up by cost of sales while 40% was the gross profit.

$$\text{b) Operating profit margin} = \frac{\text{Operating profit/Earning before interest \& tax}}{\text{Sales}}$$

The ratio indicates ability of the firm to control its operating expenses such as distribution cost, salaries and wages, travelling, telephone and electricity charges etc. e.g a ratio of 20% means:

- i) 80% of sales relate to both operating and cost of sales expenses
- ii) 20% of sales remained as operating margin profit

$$\text{c) Net profit margin} = \frac{\text{Net profit} \times 100 \text{ (earning after tax) + interest}}{\text{Sales}}$$

This ratio indicates the ability of the firm to control financing expenses in particular interest charges e.g. Net profit margin of 10% indicate that:

- i) 90% of sales were taken up by cost of sales, operating and financing expenses
- ii) 10% remained as net profits.

2. Profitability in relation to investment

$$\text{a) Return on Investment (ROI) = } \frac{\text{Net profit} \times 100}{\text{Total asset}}$$

or return on total asset (ROTA)

The ratio indicate the return on profit from investment of Sh.1 in total assets e.g a ratio of 20% means Sh.10 of total asset generated Sh.2 of net profit.

$$\text{b) Return on equity (ROE) = } \frac{\text{Net profit} \times 100}{\text{equity}}$$

or Return on net worth (RONW)
or Return on shareholders equity (ROSE)

The ratio indicate the return of profitability for every one shilling of equity capital contributed by the shareholders e.g a ratio of 25% means one shilling of equity generates Sh.0.25 profit attributable to ordinary shareholders.

$$\text{c) Return on capital employed ROCE = } \frac{\text{Net profit} \times 100}{\text{Net Asset (Capital employed)}}$$

or Return on net asset (RONA)

This ratio indicate the returns of profitability for every one shilling of capital employed in the firm.

5. The Growth and Valuation Ratio

This ratio indicates the growth potential of the firm in addition to determining the value of the firm and investment made by various investors. They include the following:

$$a) \quad \text{Earnings per share EPS} = \frac{\text{Earnings to Ordinary shareholders}}{\text{No. of ordinary shares}}$$

This ratio indicate earnings power of the firm i.e how much earnings or profits are attributed to every share held by an investor. The higher the ratio the better the firm.

$$b) \quad \text{Earnings yield (EY)} = \frac{\text{Earnings per share} \times 100}{\text{Market price per share}}$$

The market price per share (MPS) is the price at which new shares can be bought from the stock market.

These ratios therefore indicate the returns or earnings for every one shilling invested in the firm.

$$c) \quad \text{Dividends per share (DPS)} = \frac{\text{Dividend paid}}{\text{No. of ordinary shares}}$$

- This indicates the cash dividend received for every share held by an investor. If all the earnings attributable to ordinary shareholders were paid out as dividend, then EPS = DPS.

$$d) \quad \text{Dividend Yield (DY)} = \frac{\text{Dividend per share} \times 100}{\text{Market price per share}}$$

$$\text{Or} \quad \frac{\text{Dividend paid}}{\text{Market value of equity}}$$

Where market value of equity = No. of shares x MPS

This ratio indicates the cash dividend returns for every one shilling invested in the firm.

$$e) \quad \begin{array}{l} \text{Price earnings (P/E)} \\ \text{Ratio} \end{array} = \frac{\text{Market price per share (MPS)}}{\text{Earning per share}}$$

$$\text{OR} \quad = \frac{\text{Market value of equity}}{\text{Earning to Ord. Shareholders}}$$

P/E ratio is a reciprocal of earning yield (EY). The MPS is the price at which a new share can be bought i.e investment per share. The EPS is the annual income/earnings from each share.

PE therefore indicate the payback period i.e number of years it will take to recover MPS from the annual earnings per share of the firm.

$$f) \quad \text{Dividend cover} = \frac{\text{EPS}}{\text{DPS}} = \frac{\text{Earning to ordinary shares}}{\text{Dividend paid}}$$

This indicate the number of times dividend can be paid from earnings to ordinary shareholders. The higher the DPS the lower the dividend cover and vice-versa e.g consider the following two firms X and Y

	X	Y
EPS	12/=	12/=
DPS	3/=	5/=
Dividend cover	$\frac{12}{3} = 4$	$\frac{12}{5} = 2.4 \text{ times}$

$$g) \quad \text{Dividend pay out ratio} = \frac{\text{DPS}}{\text{EPS}} \times 100 = \frac{\text{Dividend paid}}{\text{Earning to ordinary shareholder}}$$

This is the reciprocal of dividend cover. It indicates the proportion of earnings that was paid out as dividend e.g a payout ratio of 40% means 60% of earnings were retained while 40% was paid out as dividend, therefore retention ratio = 1 – dividend payout ratio

$$h) \quad \text{Book value per share} = \frac{\text{Networth Equity}}{\text{No. of ordinary shares}} \quad (\text{BVPS})$$

This is also called liquidity ratio which indicates the amount attributable to each share if the firm was liquidated and all asset sold at their book value.

The ratio is based on the residual amount which would remain after paying all liabilities from the sales proceeds of the assets.

$$i) \quad \text{Market to book value per share} = \frac{\text{MPS}}{\text{BVPS}}$$

This ratio indicates the amount of goodwill attached to the firm i.e the price in excess of the sales value of the assets of the firm. If the ratio is greater 1 (MBVPS > 1) this indicate a positive goodwill while if less than 1 a –ve goodwill.

Uses/Application of Ratios

Ratios are used in the following ways by managers in various firms.

1. Evaluating the efficiency of assets utilization to generate sales revenue i.e turnover ratio.
2. Evaluating the ability of the firm to meet its short term financial obligation as and when they fall due (liquidity ratios).
3. To carry out industrial analysis i.e compare the firm's performance with the average industrial performance of the firm with that of individual competitors in the same industry.
4. For cross sectional analysis i.e compare the performance of the firm with that of individual competitors in the same industry.
5. For trend/time series analysis i.e evaluate the performance of the firm over time.
6. To establish the extent which the assets of the firm has been financed by fixed charge capital i.e use of gearing ratio

7. To predict the bankruptcy of the firm i.e use of selected ratios to determine the overall ratio usually called Z-score. The Z-score when compared with a pre-determined acceptable a Z-score will indicate the probability of the bankruptcy of the firm in future.

Limitations of Ratios

Ratios have the following weaknesses:

1. They ignore the size of the firm being compared e.g in cross-sectional analysis, the firm being compared might be of different size, technology and product diversification.
2. Effect of inflation:
Ratio ignores the effect of inflation in performance e.g increase in sales might be due to increase in selling price caused by inflationary pressure in the economy.
3. Ratios ignore qualitative or non-quantifiable aspects of the firm e.g important assets such as corporate image, efficient management team, customer loyalty, quality of product, technological innovation etc are not captured in ratio analysis.
4. Ratios are computed only at one point in time i.e they are subject to frequent changes after computation e.g liquidity ratios will constantly change as the cash, debtors and stock level changes.

5. Monopolistic firms

It is difficult to carry out industrial and cross-sectional analysis for monopolistic firms since they do not have competitors and they are the only firms in the whole industry e.g Telkom-Kenya, East Africa Brewery etc.

6. Historical Data – Ratios are computed in historical information or financial statement thus may be irrelevant in future decision-making of

7. Computation and interpretation

Generally some ratios do not have an acceptable standard of computation. This may differ from one industry to another. E.g the return on investment may be computed as:

$$\text{Return on investment} = \frac{\text{EBIT}}{\text{Total assets}} \quad \text{or} \quad \frac{\text{EAT}}{\text{Total assets}}$$

8. Different accounting policies – Different firms in the same industry use different accounting policies e.g methods of depreciation and stock valuation. This makes comparison difficult.

PRACTICE QUESTION

TRADEWINDS MANUFACTURING CORPORATION December 31, 20X2, and December 31, 20X1 Balance Sheet

	20X2	20X1
Assets		
Current assets:		
Cash	Kshs 500,000	Kshs 500,000
Marketable securities (at cost)	500,000	450,000
Accounts receivable less allowance for bad debts	2,000,000	1,600,000
Inventories	3,000,000	2,000,000
Total current assets	Kshs 6,000,000	Kshs 4,550,000
Fixed assets (property, plant, and equipment):		
Land	450,000	450,000
Building	4,000,000	4,000,000
Machinery	1,500,000	800,000
Office equipment	50,000	50,000
Less: Accumulated depreciation	2,000,000	1,700,000
Net fixed assets	4,000,000	3,600,000
Prepayments and deferred charges	400,000	300,000
Intangibles	100,000	100,000
Total assets	Kshs10,500,000	Kshs8,550,000
Liabilities		
Current Liabilities:		
Accounts payable	Kshs 1,000,000	Kshs 750,000
Notes payable	1,500,000	500,000
Accrued expenses payable	250,000	225,000
Taxes payable	250,000	225,000
Total current liabilities	Kshs 3,000,000	Kshs 1,700,000
Long term liabilities:		
First mortgage bonds, 5% interest, due 2025	3,000,000	3,000,000
Deferred taxes	600,000	600,000
Total liabilities	Kshs 6,600,000	Kshs 5,300,000
Stockholders' Equity		
Common stock, Kshs5 par value each; authorized, issued, and outstanding 300,000 shares	Kshs 1,500,000	Kshs 1,500,000
Capital surplus	500,000	500,000
Accumulated retained earnings	1,900,000	1,250,000
Total stockholders' equity	3,900,000	3,250,000
Total liabilities and stockholders' equity	Kshs10,500,000	Kshs8,550,000

Consolidated Income Statement

	20X2	20X1
Net sales	Kshs11,500,000	Kshs10,700,000
Cost of sales and operating expenses:		
Cost of goods sold	8,200,000	7,684,000
Depreciation	300,000	275,000
Selling and administration expenses	1,400,000	1,325,000
Operating profit	1,600,000	1,416,000
Other income:		
Dividends and interest	50,000	50,000
Total income from operations	1,650,000	1,466,000
Less: Interest on bonds and other liabilities	300,000	150,000
Income before provision for income tax	1,350,000	1,316,000
Provision for income tax	610,000	600,000
Net profit	Kshs 740,000	Kshs 716,000
Dividends paid out	Kshs 90,000	Kshs 132,000
Retained earnings	Kshs 650,000	Kshs 584,000

The comparable industry financial ratios for Tradewinds are given below.

Net profit margin	7.3%
Gross profit margin	14.1%
Days in inventory	91
Days in receivables	47
Days in payables	37
Current ratio	1.8
Quick ratio	0.9
Total asset turnover	1.2

Required: How well is Trade winds performing relative to its industry? (15 Marks)

REVISION QUESTIONS

QUESTION ONE

An extract from the finance statements of Kenyango Fisheries Ltd is shown below:

	Shs.
Issued share capital:	
150,000 ordinary shares of Sh.10 each	1,500,000
fully paid	2,000,000
10% loan stock 1999	1,500,000
Share premium	<u>7,000,000</u>
Revenue Reserve	<u>12,000,000</u>
Capital employed	

The profits after 30% tax is Sh.600,000. However, interest charge has not been deducted.

Ordinary dividend payout ratio is 40%.

The current market value of ordinary shares Shs.36

Required

- a) Return on capital employed
- b) Earnings per share
- c) Price earnings ratio
- d) Book value per share
- e) Gearing ratio
- f) Market to book value per share

QUESTION TWO

The following financial statements relate to the ABC Company:

Assets	Shs.	Liabilities & Net worth	Shs.
Cash	28,500	Trade creditors	116,250
Debtors	270,000	Notes payable (9%)	54,000
Stock	<u>649,500</u>	Other current liabilities	100,500
Total current assets	948,800	Long term debt (10%)	300,000
Net fixed assets	<u>285,750</u>	Net worth	<u>663,000</u>
	<u>1,233,750</u>		<u>1,233,750</u>

Income Statement for the year ended 31 March 1995

	Shs.
Sales	1,972,500
Less cost of sales	<u>1,368,000</u>
Gross profit	604,500
Selling and administration expenses	<u>498,750</u>
	105,750
Earning before interest and tax	<u>34,500</u>
Interest expense	<u>71,250</u>
	<u>28,500</u>
Estimated taxation (40%)	<u>42,750</u>

Earnings after interest and tax

Required

- a) Calculate:
- i) Inventory turnover ratio;
 - ii) Times interest earned ratio;
 - iii) Total assets turnover;
 - iv) Net profit margin

(Note: Round your ratios to one decimal place)

- b) The ABC Company operates in an industry whose norms are as follows:

Ratio	Industry Norm
Inventory turnover	6.2 times
Times interest earned ratio	5.3 times
Total assets turnover	2.2 times
Net profit margin	3%

Required

Comment on the revelation made by the ratios you have computed in part (a) above when compared with the industry average.

QUESTION THREE

The following information has been extracted from the published accounts of Pesa Corporation Limited, a company quoted on the Nairobi Stock Exchange.

	Shs.
Net profit after tax and interest	990,000
Less: dividends for the period	<u>740,000</u>
Transfer to reserves	250,000
Accumulated reserves brought forward	<u>810,000</u>
Reserves carried forward	<u><u>1,060,000</u></u>
Share capital (Sh.10 par value)	<u><u>Sh.8,000,000</u></u>
Market price per share now	<u><u>12%</u></u>

Required

- a) What is meant by a company quoted on the Nairobi Stock Exchange?
- b) Calculate for Pesa Corporation Limited the following ratios and indicate the importance of each to Miss Hisa, a Shareholder:
- i) Earnings per share.
 - ii) Price earnings ratio
 - iii) Dividend yield
 - iv) Dividend cover

QUESTION FOUR

The executive director of Pesa Ltd has circulated the following information as part of board paper:

Pesa Ltd.

Financial Performance for the year ended 31 March:

	1999	1998
i) Return on investment	12%	10%
ii) Gross profit on sales	25%	20%
iii) Number of days credit given	30 days	45 days
iv) Administrative cost of sales	7%	10%

Required

a) Brief report on each of the above 4 ratios indicating the reservation, if any, you may have or judging them as improvement in performance.

b) Tajiri Ltd has sales of Sh.20,000,000 in 1998. Beginning and closing stock was Sh.800,000 and Sh.2,200,000 respectively. G.P. margin is usually 25% of sales.

Required

- i) Stock turnover ratio
- ii) Number of days stock held
- iii) Brief explanation on how the ratio computed in (i) above can be improved and financial consequences of such action.

QUESTION FIVE

The following financial statements relate to the KCE Ltd:

KCE Ltd.
Statement of Financial Position
As at 31st March, 2015

Assets	Kshs.	Liabilities & Net worth	Kshs.
Cash	28,500	Trade creditors	116,250
Debtors	270,000	Notes payable (9%)	54,000
Stock	<u>649,500</u>	Other current liabilities	100,500
Total current assets	948,800	Long term debt (10%)	300,000
Net fixed assets	<u>285,750</u>	Net worth	<u>663,000</u>

<u>1,233,750</u>	<u>1,233,750</u>
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KCE Ltd.
Income Statement
For the year ended 31st March, 2015

	Kshs
Sales	1,972,500
Less cost of sales	<u>1,368,000</u>
Gross profit	604,500
Selling& administration expenses	<u>498,750</u>
Earnings before interest and tax	105,750
Interest expense	<u>34,500</u>
	71,250
Estimated taxation (40%)	<u>28,500</u>
Earnings after interest and tax	<u>42,750</u>

The KCE Company operates in an industry whose norms are as follows:

Ratio	Industry Norm
Inventory turnover	6.2 times
Times interest earned ratio	5.3 times
Total assets turnover	2.2 times
Net profit margin	3%

Prepare a report to be tabled in the next procurement committee meeting on the revelation made by the inventory turnover, times interest earned, total assets turnover and net profit margin ratios when compared with the industry norm (You may use appropriate computations).

(15 marks)

TOPIC 6: VALUATION CONCEPTS IN FINANCE

CONTENTS

- Bases and importance of valuation
- Valuation of ordinary shares
- Valuation of bonds and Debentures

VALUATION OF BUSINESS

A business may be valued for different reasons such as for merger, takeover, acquisition, or outright sale or liquidation. In purchasing a business, a buyer will be interested in not only the assets but also the future income this business is expected to generate.

BASES OF VALUATION

1. **Theoretical value** – In theory, if a purchaser buys a business, he is simply buying a stream of future income flows and to arrive at the actual purchase price the buyer will:
 - a) Consider the estimated probable cash flows.
 - b) Discount cash flows to their present value.
 - c) Add together the separate amounts to give the present value of income stream. Where future income flows are constant:

$$PV = C \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

Where: PV = Present value of income stream
c = Inflow per annum
r = Discounting rate
n = Number of years the inflows will last

Example

As a result of the purchase of an asset, the income stream will increase by £1,000 per annum for 25 years. Assuming a discount rate of 20%, compute the maximum price to be paid for this asset ignoring taxation.

Solution

Maximum price = Present value of all future cash inflows

$$\begin{aligned} \text{Maximum price} &= £10,000 \times PVA_{20\%, 25} \\ &= £10,000 \times \frac{1 - (1.2)^{-25}}{0.20} = 10,000 \times 4.9476 \\ &= £49,476 \end{aligned}$$

In practice the income streams are never uniform and have to be estimated from existing income shown in the recent accounts.

2. **Earning method** – The business is valued according to the total stream of income it is expected to generate over its lifetime.

Determination of maintainable earnings

- a) The first step in arriving at earning based valuation is to estimate the future maintainable earnings and if the conditions in the future are expected to be similar to those in the past, it is then prudent to base the forecast on the historical figures. However, conditions do change and as such changes in cost and revenue. Therefore, a detailed examination of profits of the most recent profit and loss account will be necessary to estimate the effects of the changes. While the information given will depend upon the nature of the business the general principles to bear in mind must include the trend of sales and gross profit.
- b) Analysis of sales and gross profit percentage by:
 - i) Product lines
 - ii) Departments
 - iii) Geographical areas
 - iv) Customer type.
- c) Costs as a percentage of total sales.
- d) Unusual fluctuations in the ratios.
- e) Necessity of expenditure in the business e.g. excessive remuneration on expenses charged.
- f) Inclusion of all costs.
- g) Effects of external conditions such as inflation or recession.

However, there are several ways of arriving at the value based on the earnings valuation.

- i) Earnings yield valuation
- ii) Price earnings ratio valuation
- iii) Super profits valuation

I) Earnings Yield Valuation

EY is given by the earnings made by the business expressed as a percentage of the market price of the business i.e.

$$EY = \frac{\text{Earnings}}{\text{Market price of equity}} \times 100$$

$$EY = \frac{EPS \times 100}{MPS} = \frac{\text{Earnings to Shareholders}}{\text{Market value of equity}}$$

$$\text{Therefore Market Value} = \frac{\text{Earning to shareholders}}{\text{Earnings yield}}$$

Example

Estimated maintainable earnings are £240,000 per annum, rate of return required is 25%.

Compute the value of the business.

$$\begin{aligned}\text{Value MV)} &= \frac{E}{EY} \times 100 \\ &= \frac{240,000}{0.25} \times 100 \\ \text{M.V.} &= £960,000\end{aligned}$$

This method can be converted into the theoretical base, especially if the business is going concern.

$$PV = \frac{C}{i} \left[1 - \frac{1}{(1+0.25)^N} \right]$$

Note

As N approaches ∞

$$\begin{aligned}Pv &= \frac{C}{r} \\ &= \frac{240,000}{0.25} = £960,000\end{aligned}$$

ii) Price Earning Ratio Valuation

P/E ratio is traditionally used for valuation of shares but it is an important ratio in the valuation of business. The P/E ratio is the measure of how many years earnings would 'purchase' the market value of the business and is given by:

$$\text{P/E ratio} = \frac{MV}{E}$$

$$MV = P/E \times E$$

NB: The value of the business can be calculated by taking estimated earnings x P/E ratio.

VALUATION OF SECURITIES:

The previous methods were ideal for valuing the entire business but it is also necessary to ascertain the value of part of a business namely shares, or securities or a block of shares in a limited liability company. The valuation of securities and shares in particular is necessary in the following aspects:

- i) To facilitate take-over bids
- ii) To allow for mergers.
- iii) To facilitate for company accounts disclosure

- iv) For purposes of acquisitions or disposal of blocks of shares.
- v) For purposes of computing capital gains tax (not applicable in Kenya at present)
- vi) For tax payers executors in assessing the capital transfers processes
- vii) For ascertaining stamp duty payable.

However, a number of parties are interested in the value of shares and securities and such will include:

- a) Company shareholders, directors and vendors of the company.
- b) The existing and prospective shareholders.
- c) Buyers of a company.
- d) Transferee and transferor parties, in particular from the point of view of income tax.
- e) Income tax department.

In this valuation, it is necessary to look at a company form:

- i) Quoted company (quoted shares)
- ii) Unquoted company (unquoted shares)

The valuation of shares will also be influenced by ownership of the company. If a company is owned by majority shareholders, its valuation will be different from if it was owned by minority shareholders. In addition, it is necessary to value shares because of:

- a) It is a requirement of the Company's Act 1948 in respect of quoted investments which should state the investment book value, market value and stock exchange value where this differs from market value. In this case, the Act recognises the fact that the value of shares may not always be reflected in the stock exchange price and for disclosure purposes, it must be reflected.
- i) In respect of unquoted investments the company must state aggregate amount of the book value and also state either the directors valuation which could be different from investors own valuation. Also the company should give specifications of the earnings and dividends attributed to these shares. These are necessary to enable interested parties to make their own valuations.
- ii) In respect of both quoted and unquoted, shares the company should give details of the shares so that they can assist in making a valuation of those shares judged to be significant for owning the company, namely, if individual investments exceed 10% of the issued shares of a given class or where the book value of the investment exceeds 10% of the company's assets.
- b) Capital transfer reasons i.e. the capital transfer requires a valuation of shares whether from one person to another or even if they are transferred at the time of death. Valuation date is important for valuation of companies' properties.

The main difficulties in valuation of shares are:

- i) Existence and method of valuation of goodwill.
- ii) Succession of company's management

- iii) Growth in dividend
- iv) Growth in equity.

BASES OF SHARE VALUATION

Share valuation can be done on the basis of income and asset values. However, on the basis of income a share will be entitled to two forms of income. For this reason the bases of valuing shares are:

- i) Earnings method
- ii) Dividend method
- iii) Assets method

i) Earnings Method (Or Earning Basis Valuation)

Using the earning valuation method, a company will use its P/E ratio to value its shares.

$$P/E = \frac{MV}{E}$$

$$MV = E \times P/E \rightarrow \text{value of ordinary share.}$$

The MV can be determined where the estimated earnings have been established by applying the P/E ratio expected of this type of company.

Example

Company XYZ is expected to generate post tax earnings of Sh.200,000 per annum and companies in the same trade will generally have a P/E ratio of eight (8). On account of company XYZ limited size, a ratio of six (6) is considered more appropriate. The issued share capital is 1,000,000 ordinary shares of Sh.50 each.

Required

$$\begin{aligned} \text{Value of shares} &= \text{EPS} \times P/E \\ &= \text{Earnings per share} \times P/E \\ &= \frac{200,000}{1,000,000} \times 6 = \text{Sh.12.00} \end{aligned}$$

$$\text{Value of Business} = \text{Earnings} \times P/E \text{ ratio}$$

$$MV = E \times P/E = \text{Sh.200,000} \times 6 = \text{Sh.1.2 million}$$

ii) Dividend Basis Valuation

Ownership of shares in entities – The owner to receive a cash flow consisting of future dividends and the value of a share should correspond to the present value of this future cash flow. A shareholder cannot expect cash flows in perpetuity as he will sell his shares at one time.

$$P_0 = \frac{D_0}{K_e}$$

Note: Where there is growth in equity, $P_0 = \frac{d_0(1+g)}{K_e - g}$

Example 1

Company XYZ pays a dividend of 10% on its Sh.60 par value ordinary shares. This company uses a discount rate of 15%. Assuming no growth, compute the value of its ordinary share if there's growth of 5%, what would be the value of this company's ordinary shares.

a) $P_0 = \frac{D_0}{K_e} \quad P_0 = \frac{6}{15\%} = \text{Sh.40} \quad (\text{no growth})$

b) $P_0 = \frac{6(1.05)}{0.15-0.05} = \text{Shs.63} \quad (5\% \text{ growth rate})$

EXAMPLE 2

The most recent data for Safariloam Ltd disclose the following

Dividend per share	KShs.3.00
Expected annual dividend growth rate	6 %
The current required rate of return	15 %

The company is considering a variety of proposals in order to redirect the firm's activities. The following four alternative strategies have been suggested:-

- (i) Do nothing in which case key financial variables will remain unchanged.
- (ii) Invest in a venture that will increase the dividend growth rate to 7% and increase the required rate of return to 14 %
- (iii) Eliminate any unprofitable product line. The action will increase the dividend growth rate to 8 % and raise the required rate of return to 17 %
- (iv) Acquire a subsidiary operation from another company. This action will increase the dividend growth rate to 9 % and the required rate of return to 18 %.

For each of the proposed actions determine the resulting market prices per share and recommend the best alternative strategy

iii) Asset Based Valuation

This method takes into account the entire business with reference to its assets and then divides the resultant value by the number of shares in an issue to give the per share. The principles are the same as those in the valuation of businesses computed already. However, if a historical dividend based on earning based valuation produces a figure which is less than the asset value then there is a possibility that the buyer may be able to improve the management of the asset being taken over. In such a case, a buyer would be prepared to pay a price which though excessive in terms of income might be justified by the underlying assets value.

Example 1

Information extracted from the books of Kent Limited.

	Sh.		Sh.
Current liabilities	300,000	Land	250,000
Bank overdraft	<u>50,000</u>	Stock in trade	<u>100,000</u>
	<u>350,000</u>		<u>350,000</u>

Stock has a realizable value of Sh.80,000 and land Sh.300,000. This company is assumed to be have a share capital of 20,000 ordinary shares.

Compute the value of its shares.

i) Assets method

Assets = L & B	300,000	
Stock	<u>80,000</u>	
	380,000	
Liabilities	<u>[350,000]</u>	
	<u>30,000</u>	
Value of shares	= $\frac{30,000}{20,000}$ =	Sh.1.50

Example 2

K & K Company Limited is planning to absorb three other companies so as to realise its sales records of Sh.500,000 per annum. Its accountants have advised the company to maintain such a size that it will enable its shares to sell at a minimum price of Sh.16. The company's last published balance sheets indicate the following:

	Sh.'000'	
Ordinary shares of Sh.10 each	50,000	
Reserves	65,000	
Current liabilities	<u>40,000</u>	
Total	<u>155,000</u>	
Assets:	Sh.	
Fixed assets	80,000	
Current assets	<u>75,000</u>	
Total	<u>155,000</u>	

Profits for the last 5 years were as follows:

	Sh.'000'
1.	9,000
2.	6,000
3.	10,000
4.	8,000
5.	17,000

Compute the value of the business indicating the lowest offer price and the highest offer price and the share value thereof whether it would be viable to take on the three companies if its to maintain this share value.

P/E = 12:1 Average profits = 10,000,000

$$\text{Value of shares} = \frac{\text{Sh.120 million}}{5 \text{ million shares}} = \text{Sh.24}$$

	Sh.'000'
Assets	155,000
Less: Current liabilities	[40,000]
	<u>115,000</u>

Where: P_0 = Price of ordinary shares
 d = Dividend at the end of year one
 P_1 = Price of the share at the end of one year.

This will depend on expected cash flows consisting of annual interest plus the principal amount to be received at maturity. The appropriate rate of capitalization or discount rate to be applied will depend upon the riskiness of the bond e.g. government bonds are less risky and will therefore call for lower discount rates than similar bonds issued by private companies which will call for high rate of discount.

When a bond or debenture has reached maturity, its value can be determined by considering annual interest payments plus its terminal or maturity and this is done using the P.V. concept to discount the cash flows and the result will be compared to the market value of the bond to ascertain whether it has overvalued or undervalued.

Where: Int = Annual interest
Kd = Required rate of return
M = Terminal/maturity value
n = Number of years to maturity

Example

K is contemplating purchasing a 3 year bond worth 40,000/= carrying a nominal coupon rate of interest of 10%. K required rate of return is 6%.

What should he be willing to pay now to purchase the bond if it matures at par?

Solution

$$\begin{aligned}\text{Int} &= 10\% \times 40,000 = 4,000 \text{ p.a.} \\ n &= 3 \text{ yrs} \\ K_d &= 6\% \\ M &= 40,000\end{aligned}$$

$$\begin{aligned}V_d &= \frac{4,000}{(1.06)^1} + \frac{4,000}{(1.06)^2} + \frac{4,000}{(1.06)^3} + \frac{40,000}{(1.06)^3} \\ &= 4,000 \times PVA_{6\%,3} + 40,000 \times PVIF_{6\%,3} = (40,000 \times 2.673) + (40,000 \times 0.840) = 44,292\end{aligned}$$

REVIEW QUESTIONS

QUESTION ONE

- XYZ Ltd is expected to pay a DPS of Sh.6 in one year's time. The dividend payout ratio is 60% and the Return on Equity is 15%. Determine whether the share is overvalued if the MPS is Sh.40.
- What is the significance of valuation securities?
- ABC Ltd has issued a 5 year zero coupon rate bond with maturity value of Sh.100,000. The bond is issued at a discount of 32%. Determine the rate of return of the bond.
- What are the advantages of zero coupon bond?

QUESTION TWO

Nyakua Limited is contemplating acquiring Uza Limited.

Incremental cash flows arising from the acquisition are expected to be as follows:

	Average of years (in Sh.'000')		
	1-5	6-10	11-∞
Cash flow after taxes	100	150	200
Investment required	<u>50</u>	<u>60</u>	<u>70</u>
Net cash flow	<u>50</u>	<u>90</u>	<u>130</u>

Uza Limited has an all equity capital structure. The required rate of return of Uza Limited is always 5 percent above the risk free rate. The risk free rate is 9 percent.

Required

- Using the information provided, compute the maximum price that Nyakua Limited might pay for Uza Limited.

- b) What other factors might influence the management of Nyakua Limited in their decision to purchase Uza Limited?

QUESTION THREE

- a) Andreas Company Ltd. currently pays a dividend of Sh.2 per share and this dividend is expected to grow at an annual rate of 15% for the first 3 years then at a rate of 10% for the next 3 years after which it is expected to grow at a rate of 5% thereafter.
What value would you place on the stock if an 18% rate of return were required?
Would your valuation change if you expected to hold the stock for only 3 years? Explain.

- a) The stream of dividends of XYZ Ltd for the past 4 years was as follows:

Year	1999	2000	2001	2002
DPS Sh.	2.50	2.65	2.76	2.81

The cost of equity is 14%. Determine the price of a share.

QUESTION FOUR

- a) The valuation of ordinary shares is more complicated than the valuation of bonds and preference shares. Explain the factors that complicate the valuation of ordinary shares.

The most recent financial data for the Rare Watts disclose the following:

Dividend per share	Sh.3.00
Expected annual dividend growth rate	6 percent
Current required rate of return	15 percent

The company is considering a variety of proposals in order to redirect the firm's activities. The following four alternatives have been suggested:

1. Do nothing in which case the key financial variables will remain unchanged.
2. Invest in venture that will increase the dividend growth rate to 7% and lower the required rate of return to 14%.
3. Eliminate an unprofitable product line. The action will increase the dividend growth rate to 8% and raise the required rate of return to 17%.
4. Acquire a subsidiary operation from another company. This action will increase the dividend growth rate to 9% and required rate of return to 18%.

Required

For each of the proposed actions, determine the resulting impact price and recommend the best alternative.

TOPIC 7: SOURCES OF BUSINESS FINANCE

1. Equity capital
2. Debt finance
3. Bills of exchange
4. Lease finance
5. Overdraft finance
6. Plastic money – Debenture finance
7. Venture capital

1. EQUITY FINANCE

For small companies, this is personal savings (contribution of owners to the company). For large companies equity finance is made of ordinary share capital and reserves; (both revenue and capital reserves). Equity finance is divided into the following classes:

- a) **Ordinary share capital** – this is raised from the public from the sale of ordinary shares to the shareholders. This finance is available to limited companies. It is a permanent finance as the owner/shareholder cannot recall this money except under liquidation. It is thus a base on which other finances are raised.

Ordinary share capital carries a return that is variable (ordinary dividends). These shares carry voting rights and can influence the company's decision making process at the AGM.

These shares carry the highest risk in the company (high securities – documentary claim to) because of:

- a) Uncertainty of return
- b) Cannot ensure refund
- c) Have residual claims – claim last on profits, claim last on assets.

However this investment grows through retention.

Rights of ordinary shareholders

1. Right to vote
elect BOD
 - a. Sales/purchase of assets
2. Influence decisions:
 - a) Right to residual assets claim
 - b) Right to amend company's by-laws
 - c) Right to appoint another auditor
 - d) Right to approve merger acquisition
 - e) Right to approve payment of dividends

Reasons why ordinary share capital is attractive despite being risky

- Shares are used as securities for loans (a compromise of the market price of a share).

- Its value grows.
- They are transferable at capital gain.
- They influence the company's decisions.
- Carry variable returns – is good under high profit
- Perpetual investment – thus a perpetual return
- Such shares are used as guarantees for credibility.

Advantages of using ordinary share capital in financing

- They facilitate projects especially long-term projects because they are permanent..
- Its cost is not a legal obligation.
- It lowers gearing level – reduces chances of receivership/liquidation.
- Used with flexibility – without preconditions.
- Such finances boost the company's credibility and credit rating.
- Owners contribute valuable ideas to the company's operations (during AGM by professionals).

b) RETAINED EARNINGS

i) Revenue Reserves

These are undistributed earnings. Such reserves are retained for the following reasons:

- To make up for the fall in profits so as to sustain acceptable risks.
- To sustain growth through plough backs. They are cheap source of finance.
- They are used to boost the company's credit rating so they enable further finance to be obtained.
- It lowers the company's gearing ratio – reduces chances of receivership/liquidation.

ii) Capital Reserves

1. It is raised by selling shares at a premium. (The difference between the market price (less floatation costs) and par value is credited to the capital reserve).
2. Through revaluation of the company's assets. This leads to a fictitious entry which is of the nature of a capital reserve.
3. By creation of a sinking fund.

c) PREFERENCE SHARE CAPITAL (Quasi-Equity)

It is also called quasi-equity because it combines features of equity and those of debt. It is preference because it is preferred to ordinary share capital that is:

- i) It is paid dividends first – preferred to dividend
- ii) It is paid asset proceeds first – preferred to assets.

Unlike ordinary share capital, it has a fixed return. It carries no voting rights. It is an unsecured finance and it increases the company's gearing ratio.

CLASSIFICATION

i) Redeemable Class

Redeemable preferential shares are bought back by issuing company after minimum redemption period but before expiring of maximum redemption period after which they become creditors. (Can sue the company).

ii) Irredeemable Preference Shares

Are perpetual preference shares as they will not be redeemed in the company's lifetime unless it is under liquidation, (it is permanent).

vi) Non-Cumulative Preference Shares

These cannot claim interest in arrears.

vii) Convertible

These can be converted into ordinary shares (which is optional).

Conversion ratio = par value of ordinary share/par value of preference shares e.g if par value of ordinary shares is Sh.10 and that of preference shares is Sh.20, then conversion ratio = $\frac{10}{20} = \frac{1}{2}$ i.e for every preference share you get 2 ordinary shares.

Conversion price par value of preference shares/no. of ordinary shares to be acquired.

$$= \frac{20}{2} = \text{Shs}10$$

Example

Company XYZ Ltd has sold 10,000 ordinary shares of Shs.30 (partly called up) plus 20,000 Shs.45 preference shares, which are convertible. Compute the total number of ordinary shares after conversion.

Solution

Conversion ratio = $30/45 = 2/3$ for every 2 preference shares you get 3 ordinary shares.

$$\frac{20,000}{1} \times \frac{3}{2} = 30,000 \text{ ordinary shares.}$$

$$\text{Conversion price} = \frac{45}{3/2} = \text{Shs}30$$

Total = 40,000 ordinary shares after conversion.

viii) Non-Convertible Preference Shares.

These cannot be converted into ordinary shares.

2. DEBT FINANCE

Debt finance is a fixed return finance as the cost (interest) is fixed on the par value (face value of debt). It is ideal to use if there's a strong equity base. It is raised from external sources to qualifying companies and is available in limited quantities. It is limited to:

- i) Value of security.
- ii) Liquidity situation in a given country. It is ideal for companies where gearing allows them to raise more debt and thus gearing level.

Classification of Debt Finance

Loan finance – this is a common type of debt and is available in different terms usually short term. Medium term loans vary from 2 - 5 years. Long-term loans vary from 6 years and above

The terms are relative and depend on the borrower. This finance is used on the basis of Matching approach i.e. matching the economic life of the project to the term of the loan. It is prudent to use short-term loans for short-term ventures i.e. if a venture is to last 4 years generating returns, it is prudent to raise a loan of 4 years maturity period.

Conditions under Which Loans Are Ideal

- a) When the company's gearing level is low (the level of outstanding loans is low.
- b) The company's future cash flows (inflows and their stability) must be assured. The company must be able to repay the principal and the interest.
- c) Economic conditions prevailing. The company must have a long-term forecast of the prevailing economic condition. Boom conditions are ideal for debt.
- d) When the company's market share guarantees stable sales.
- e) When the company's anticipated future expansion programs, justify such borrowing.

Requirements for Raising Loan

- a) History of the company and its subsidiaries.
- b) Names, ages, and qualifications of the company's directors.
- c) The names of major shareholders – 51% plus i.e. owner who must give consent.
- d) Nature of the products and product lines.
- e) Publicity of the product.
- f) Nature of the loan – either secured, floating or unsecured.
- g) Cash flow forecast.

Reasons Why Commercial Banks Prefer To Lend Short Term Loans

- a) Long-term forecasts are not only difficult but also vague as uncertainties tend to jeopardize planning e.g. political and economic factors.
- b) Commercial banks are limited by the Central Bank of Kenya in their long term lending due to liquidity considerations.
- c) Short-term loans are profitable. This is because interest is high as in overdrafts.
- d) Long term finance loses value with time due to inflation.
- e) Cost of finance – in the long term, the cost of finance may increase and yet they cannot pass such a cost to borrowers since the interest rate is fixed.
- f) Commercial banks do credit analysis that is limited to short term situations.

- g) Usually security market favors short term loans because there are very few long term securities and as such commercial banks prefer to lend short term due to security problems.

Advantages of Using Debt Finance

- Interest on debt is a tax allowable expense and as such it is reduced by the tax allowance.

Example

Interest = 10% tax rate = 30%

$$\begin{aligned}\text{The effective cost of debt (interest)} &= \text{Interest rate } (1 - T) \\ &= 10\% (1 - 0.30) \\ &= 7\%\end{aligned}$$

Consider companies A and B

Company	A	B
	Sh.'000'	Sh.'000'
10% debt	1,000	-
Equity	<u>-</u>	<u>1,000</u>
	<u>1,000</u>	<u>1,000</u>

The tax rate is 30% and earnings before interest and tax amount to Ksh.400, 000. All earnings are paid out as dividends. Compute payable by each firm.

Company	A	B
	Sh.'000'	Sh.'000'
EBIT	400	400
Less interest 10% x 1,000	<u>(100)</u>	<u>-</u>
EBT	300	400
Less tax @ 30%	<u>(90)</u>	<u>(120)</u>
Dividends payable	<u>210</u>	<u>280</u>

Company A saves tax equal to Sh.30, 000(120,000 – 90,000) since interest charges are tax allowable and reduce taxable income.

- The cost of debt is fixed regardless of profits made and as such under conditions of high profits the cost of debt will be lower.
- It does not call for a lot of formalities to raise and as such it is ideal for urgent ventures
- It is usually self-sustaining in that the asset acquired is used to pay for its cost i.e. leaving the company with the value of the asset.
- In case of long-term debt, amount of loan declines with time and repayments reduce its burden to the borrower.
- Debt finance does not influence the company's decision since lenders don't participate at the AGM.

Disadvantages

- It is a conditional finance i.e. it is not invested without the approval of lender.

- Debt finance, if used in excess may interrupt the companies decision making process when gearing level is high, creditors will demand a say in the company i.e. and demand representation in the BOD.
- It is dangerous to use in a recession as such a condition may force the company into receivership through lack of funds to service the loan.
- It calls for securities which are highly negotiable or marketable thus limiting its availability.
- It is only available for specific ventures and for a short term, which reduces its investment in strategic ventures.
- The use of debt finance may lower the value of a share if used excessively. It increases financial risk and required rate of return by shareholders thus reduce the value of shares.

Differences between Debt Finance and Ordinary Share Capital (Equity Finance)

Ordinary share capital	Debt
a) It is a permanent finance	a) It is refundable (redeemable)
b) Return paid when available	b) It is fixed return capital
c) Dividends are not tax allowable	c) Interest on debt is a tax allowable expense
d) Unsecured finance	d) Secured finance
e) Carry voting rights	f) No voting right
f) Reduces gearing ratio	g) Increases gearing ratio
g) No legal obligation to pay	h) A legal obligation to pay
h) Has a residue claim	i) Carries a superior claim
i) Owners' money	Creditors finance.

Similarities between Preference and Equity Finance

- a) Both may be permanent if preference share capital is irredeemable (convertible).
- b) Both are naked or unsecured finances.
- c) Both are traded at the stock exchange
- d) Both are raised by public limited companies only
- e) Both carry residue claims after debt.
- f) Both dividends are not a legal obligation for the company to pay.

Differences between Preference and Equity Finance

Ordinary share capital	Preference share capital
a) Has a residue claim both on assets and profit	a) Has a superior claim
b) Carries voting rights	b) No voting rights
c) Reduces the gearing ratio	c) Increases the gearing ratio
d) Variable dividends hence grow over time	d) Fixed dividends hence no growth
e) Permanent finance	e) Usually redeemable
f) Easily transferable.	f) Not easily transferable

Similarities between Debt and Preference Share Capital

- a) Both have fixed returns.
- b) Both will increase the company's gearing ratio.
- c) Both are usually redeemable.
- d) Both do not have voting rights.
- e) Both may force the company into receivership
- f) Both have superior claims over and above owners.
- g) Both are external finances.
- h) There is no growth with time.

Differences between Preference Share Capital and Debt

DEBT	PREFERENCE SHARE CAPITAL
a) Interest is tax allowable	a) Dividends are not tax allowable
b) Interest is a legal obligation	b) Dividends are not a legal obligation
c) Debt finance is always secured	c) Preference is not secured finance
d) Debt finance is a pre-conditional	d) Is not conditional finance
e) Has a superior claim	e) Has a residue claim (after debt)

Why It May Be Difficult For Small Companies to Raise Debt Finance in Kenya (Say Jua Kali Companies)

- Lack of security
- Ignorance of finances available
- Most of them are risky businesses as there are no feasibility studies done (chances of failure have been put to 80%).
- Their size being small tends to make them UNKNOWN i.e. they are not a significant competitor to the big companies.
- Cost of finance may be high – their market share may not allow them to secure debt.
- Small loans are expensive to extend by bank i.e. administration costs are very high.
- Lack of business principles that are sound and difficult in evaluating their performance.

Solutions to the Above Problems

- There should be diversification of securities e.g. to accept guarantees.
- Education of such businessmen on sound business principles.
- The government should set up a special fund to assist the Jua kali businessmen.
- Encourage formation of co-operative societies.
- To request bankers to follow up the use of these loans.

3. Bills of Exchange

Bills of Exchange are a source of finance in particular in the export trade. A Bill of Exchange is an unconditional order in writing addressed by one person to another

requiring the person to whom it is addressed to pay to him as his order a specific sum of money. The commonest types of bills of exchange used in financing are accommodation bills of exchange. For a bill to be a legal document; it must be

- a) Drawn by the drawer.
- b) Bear a stamp duty
- c) Acceptable by the drawee
- e) Mature in time.

It is used to raise finance through:

- i) Discounting it.
- ii) Negotiating
- iii) Giving it out as security.

Advantages of Using a Bill as a Source of Finance

- They are a faster means of raising finance (if drawer is credible).
- Is highly negotiable/liquid investment
- Does not require security
- Does not affect the gearing level of the company
- It is unconditional and can be invested flexibly
- It is useful as a source of finance to finance working capital
- It is used without diluting capital.

4 Lease Finance

Leasing is a contract between one party called lessor (owner of asset) and another called lessee where the lessee is given the right to use the asset (without legal ownership) and undertakes to pay the lessor periodic lease rental charges due to generation of economic benefits from use of the assets. Leases can be short term (operating leases) in which case the lessor incurs the operating and maintenance costs of the assets or long term (finance leases) in which the lessee maintains and insure the assets.

Lease finance is ideal under the following conditions:

- a) When the asset depreciates faster.
- b) When the asset is subject to obsolescence
- c) When the available asset cannot meet the contemplated expansion program
- d) When the asset's cost is prohibiting
- e) If the asset is required seasonally
- f) If the asset can generate returns to pay off lease charges in the short run.

Advantage of Leasing an Asset

- It does not tie up the company's funds in an asset.
- The arrangement may ensure lessor bears the maintenance costs reducing the companies operating costs.
- The company has the option to purchase assets at the expiry of the lease period at which time it will know the viability of the asset.

- The company (lessee) will enjoy the lease charges as allowable expenses thus reducing taxable income and tax liability.
- Lease finance enables the lessee to use the asset to create financial surpluses which may then be used to buy assets.
- It is usually a long-term arrangement which enables the company to plan returns expected and operations which may be carried out.

Disadvantage of Leasing an Asset

- It is a pre-conditional finance (as on the use of asset)
- In the long term the lease charges may out-weigh the cost of buying own asset.
- It is available for a selected asset and this limits flexibility.
- It is useful for financing fixed assets and not working capital
- Lease finance may not be renewed leading to loss of business.
- Lease financing lowers the company's credit rating (i.e. the asset in the balance sheet is shown as leased asset).

Reasons Why Lease Finance Is Not Well Developed In Kenya

- Lease charges are usually prohibitive i.e. the cost of finance is excessive.
- It may not be known to businessmen.
- Uncertainty as to returns from such assets i.e. the returns from such assets leased may not encourage the growth of lease finance.
- There is an imperfect market as a number of companies lease assets on basis of credibility of the lessee.
- Lack of flexibility i.e. a number of assets which are ideal for leasing are unavailable.
- Kenya's financial markets are underdeveloped and this has affected the development of lease finance.
- After lease service is poor and this leads to loss of revenue.

5. Overdraft Finance

This finance is ideal to use as bridging finance in sense that it should be used to solve the company's short term liquidity problems in particular those of financing working capital (w.c.). It is usually a secured finance unless otherwise mentioned. Overdraft finance is an expensive source of finance and the over-reliance on it is a sign of financial imprudence as it indicates the inability to plan or forecast financial needs.

Advantages of Overdraft Finance

- It is useful in financial crisis which an accountant cannot forecast due to abrupt fall in profits thus liquidity problems.
- In some cases it may be secured on goodwill thus making it flexible finance.
- It does not entail preconditions and is therefore investible in high-risk situations when the firm would not have finance in normal circumstances.
- It is raised faster and as usual is ideal to invest in urgent ventures e.g. documentary investments e.g. treasury bonds, shares, treasury bills, housing bonds etc.

- If not used for a long period of time – it does not affect the company's gearing level and therefore does not relate to company's liquidation or receivership.
- Less formalities/procedures involved.

Disadvantages of Overdraft Finance

- It is expensive as the interest rates of overdrafts are much higher than bank rates.
- The use of this finance is an indication of poor financial management principle.
- It may be misused by management because it does not carry pre-conditions
- Being a short-term financial arrangement, it can be recalled at short notice leaving the company in financial crisis.

6. Plastic Money (Credit Card Finance)

This is finance of a kind whereby a company will make arrangements for the use of the services of a credit card organization (through the purchase of credit cards) in return for prompt settlement of bills on the card and a commission payable on all credit transactions. This is used to finance goods and services of working capital in nature such as the payment of fuel, spare- parts, medical and other general provisions and it is rare for it to finance raw materials or capital items.

Reasons behind the Fast Development of This Finance (Plastic Money) In Kenya

- a) High incidences of fraud by dishonest employees has been responsible for development of this finance as it minimizes chances of this fraud because it eliminates the use of hard cash in the execution of transactions.
- b) Risk associated with carrying of huge amounts of cash for purchases which cash is open to theft and misuse has also been responsible for development of this finance.
- c) Credit cards have boosted the credibility of holder companies which enables them to obtain trade credits under conditions which would have otherwise been difficult.
- d) Of late, Kenya has experienced emergence of elite, middle and high-income groups' in particular professionals who tend to use these cards as a symbol of status in execution of day to day transactions.
- e) These cards have been used by financial institutions and banks to boost their deposit and attract long term clienteles e.g. Royal Card Finance, Standard Chartered.
- f) A number of companies and establishments have acquired such cards as a means of settling their bills under certain times when their liquidity is low or when in financial crisis.

Limitations of Credit Cards as a Source of Finance

- i) These cards lead to overspending on the part of the holder and as such may disorganize the organization's cash budget and cash planning.
- ii) Limited as to the activities they can finance as they are ideal for financing working capital items and not fixed assets in which case they are not a profitable source of finance.
- iii) They are expensive to obtain and maintain because of associated cost such as ledger fees, registration, insurance, commission expenses, renewal fees etc.

- iv) It is a short-term source and is open only to a few establishments in which case a company can obtain goods and services from those establishments that can accept them.
- v) Entail a lot of formalities to obtain e.g. guarantees, presentation of bank statements and even charging assets that are partially pledged to secure expenses that may be incurred using these cards.
- vi) They may be misused by dishonest employees who may use them to defraud the organization off goods and services which may not benefit such organizations.
- vii) Credit card organization may suspend the use of such cards without notice and this will inconvenience the holder who may not meet his/her ordinary needs obtained through these cards.

7. Debenture Finance

A form of long term debt raised after a company sells debenture certificates to the holder and raises finance in return. The term debenture has its origin from 'DEBOE' which means 'I owe' and is thus a certificate or document that evidences debt of long term nature whereby the person named therein will have given the issuing company the amount usually less than the total par value of the debenture. These debentures usually mature between 10 to 15 years but may be endorsed, negotiated, discounted or given as securities for loans in which case they will have been liquidated before their maturity date. The current interest rate is payable twice a year and it is a legal obligation.

Classification

i) Secured Debentures

These are those types of debentures which a company will secure usually in two ways, secured with a fixed charge or with a floating charge.

- a) Fixed Charge – a debenture is secured with a fixed charge if it can claim on a specific asset.
- b) Floating charge – if it can claim from any or all of the assets which have not been pledged as securities for any other form of debt.

ii) Naked Debentures

These are not secured by any of the company's assets and as such they are general creditors.

iii) Redeemable Debentures

These are the type of debentures, which the company can buy back after the minimum redemption period and before the maximum redemption period (usually 15 years) after which holders can force the company to receivership to redeem their capital and interest outstanding.

iv) Irredeemable Debentures (perpetuities)

These are never bought back in which case they form permanent source of finance for the company. However, these are rare and are usually sold by companies with a history of stable ordinary dividend record.

v) Classification according to convertibility

Convertible debentures – Can be converted into ordinary shares although they can also be converted into preference shares.

Conversion price = par value of a debenture/No. of shares to be received.

$$\text{Conversion ratio} = \frac{\text{Par value of debenture}}{\text{Par value of ordinary shares}}$$

Example

ABC Company Ltd books:

	Shs.
10,000, Shs.20 ordinary share capital	200,000
10,000, Shs.10 8% preference share capital	100,000
5,000, Shs.100 12% debentures	500,000
The above debentures are due for conversion:	

Required

- i) Compute the conversion price
- ii) Compute the conversion ratio
- iii) Compute new capital structure.

Solution

- i) Conversion price = par value of debenture/No. of shares to be received.

$$\begin{aligned} \text{No. of shares to be received} &= 100:20 = 5:1 \\ \text{Therefore} &= \frac{100}{5} = 20 \end{aligned}$$

- ii) Conversion ratio = par value of debenture/par value of share = $\frac{100}{20} = 5.0$
Receive 5 ordinary shares for every 1 debenture held.

- iii) New capital structure

No. of new ordinary shares	=	5000 x 5 =	25,000
			Shs.
35,000, Shs.20 ordinary shares			
700,000			
10,000, Shs.10, 8% preference shares			
<u>100,000</u>			
Total capital			<u>800,000</u>

vi) Non-convertible debentures

These cannot be converted into ordinary preference shares and they are usually redeemable.

vii) Sub-ordinate debentures

Usually last for as long as 10 years and they are sold by financially strong companies. Such are not secured and they rank among general creditors in claiming on assets during liquidation. This means that they are sub-ordinate to senior debt but superior to ordinary and preference share capital.

Reasons behind Unpopularity of Debentures of Kenya's Financial Market

- i) Their par value is an extremely high value and as such they are unaffordable to purchase by would be investors.
- ii) They are in most cases secured debt and as such constrain the selling company in so far as getting sufficient securities is difficult.
- iii) Most of the would-be sellers have low credit worthiness which is difficult.
- iv) Kenya's capital markets are not developed and as such there is no secondary debenture market where they can be discounted or endorsed.
- v) Debentures finance is not known among the general business community and as such many would be sellers and buyers are ignorant of its existence.
- vi) Being long term finance there are a few buyers who may be willing to stake their savings for a long period of time.
- vii) Such finance calls for a fixed return, which in the long run will be eroded by inflation.

8. Venture Capital

Venture capital is a form of investment in new small risky enterprises required to get them started by specialists called venture capitalists. Venture capitalists are therefore investment specialists who raise pools of capital to fund new ventures which are likely to become public corporations in return for an ownership interest. They buy part of the stock of the company at a low price in anticipation that when the company goes public, they would sell the shares at a higher price and therefore make a considerably high profit.

Venture capitalists also provide managerial skills to the firm. Example of venture capitalists are pension funds, wealthy individuals, insurance companies, Acacia fund, Rock fella, etc.

Since the goal of venture capitalists is to make quick profits, they will invest only in firms with a potential for rapid growth.

Venture capitalists, will only invest in a company if there is a reasonable chance that the company will be successful. Their publicity material states that successful investments have three common characteristics.

- a) There is a good basic idea, a product or service which meets real customer needs.
- b) There is finance, in the right form to turn the idea into a solid business.
- c) There is the commitment and drive of an individual or group and the determination to succeed.

Attributes of venture capital

- i) Equity participation – Venture Capital participates through direct purchase of shares or fixed return securities (debentures and preference shares)

- ii) Long term investment – venture capital is an investment attitude that necessitates the venture capitalists to wait for a long time (5 – 10 years) to make large profits (capital gains).
- iii) Participation in Management – Venture capitalists give their Marketing, Planning and Management Skills to the new firm. This hands – on Management enable them protect their investment.

Role of Venture Capital in Economic Development

The types of venture that capitalists might invest will involve:

- a) **Business start-ups** – When a business has been set up by someone who has already put time and money into getting it started, the group may be willing to provide finance to enable it to get off the ground. With start-ups, venture capital often prefers to be one of several financial institutions putting in venture capital.
- b) **Business development** – The group may be willing to provide development capital for a company which wants to invest in new products or new markets or to make a business acquisition, and so which so needs a major capital injection.
- c) **Management buyout** – A management buyout is the purchase of all or parts of a business from its owners by its managers.
- d) Helping a company where one of its owners wants to **realize all or part of his investment**. The venture capital may be prepared to buy some of the company's equity.

Funding Venture Capital

When a company's directors look for help from a venture capital institution, they must recognize that:

- a) The institution will want an equity stake in the company.
- b) It will need convincing that the company can be successful (management buyouts of companies which already have a record of successful trading have been increasingly favored by venture capitalists in recent years.
- c) It may want to have a representative appointed to the company's board, to look after its interests.

The directors of the company then contract venture capital organizations, to try to find one or more which would be willing to offer finance. A venture capital organization will only give funds to a company that it believes can succeed.

Reasons for Significant Growth in Venture Capital in the Developed Countries

- i) Public attitude i.e. a favourable attitude by the public at large towards entrepreneurship, success as well as failure.
- ii) Dynamic financial system e.g. efficient stock exchange and a competitive banking system.
- iii) Government support – e.g. taxation system to encourage venture capital e.g. tax concessions and investment allowance taxes.
- iv) Establishment of venture capital institutions e.g. investors in the industry.
- v) Growth in the number of Management buyers (MBO) which have created a demand for equity finance.

Constraints of Venture Capital in Kenya

1. Lack of rich investors in Kenya, hence inadequate equity capital.
 2. Inefficiencies of stock market – NSE is inefficient and investors cannot sell the shares in future. Prices do not reflect all the available information in the market.
 3. Infrastructural problems – this limits the growth rate of small firms which need raw materials and unlimited access to the market factors of production.
 4. Lack of managerial skills on part of venture capitalists and owners of the firm.
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1. Nature of small business in Kenya. There are 3 categories.
 - a. Large MNC – these are established firms and can raise funds easily.
 - b. Asian owned small businesses – They are family owned hence do not require interference of venture capitalists because they are not ready to share profits.
 - c. African – owned business – need venture capital but have little potential for growth.
 6. Focus on low risk ventures e.g. confining to low technology, low growth sectors with minimum investment risks.
 7. Conservative approach by the venture capitalists.
 8. Delay in project evaluation e.g. months or more hence entrepreneurs loose interest in the project.
 9. Lack of government support and inefficient financial system.

Summary

In sum, venture capital, by combining risk financing with management and marketing assistance, could become an effective instrument in fostering developing countries. The experiences of developed countries and the detailed case study of venture capital however, indicate that the following elements are needed for the success of venture capital in any country.

- A broad-based (and less family based) entrepreneurial traditional societies and government encouragement for innovations, creativity and enterprise.
- A less regulated and controlled business and economic environment where attractive customer opportunities exist or could be created from high-tech and quality products.
- Existence of disinvestments mechanisms, particularly over-the counter stock exchange catering for the needs of venture capitalists.
- Fiscal incentives which render the equity investment more attractive and develops 'equity cult' in investors.
- A more general, business and entrepreneurship oriented education system where scientists and engineers have knowledge of accounting, finance and economics and accountants understand engineering or physical sciences.
- An effective management education and training programme for developing professionally competent and committed venture capital managers; they should be trained to evaluate and manage high technology, high risk ventures.
- A vigorous marketing thrust, promotional efforts and development strategy, employing new concepts such as venture fair clubs, venture networks, business incubators etc. for the growth of venture capital.

- Linkage between universities/technology institutions, R & D. Organizations', industry, and financial institutions including venture capital firms.
- Encouragement and funding of R & D by private public sector companies and the government for ensuring technological competitiveness.

Disadvantages of Venture Capital

- Dilute ownership position of a firm
- Dilute control of a firm